



MODULE 1: EXCHANGE-TRADED FUNDS: MECHANICS AND APPLICATIONS

LEARNING OUTCOMES

- 1.a.** Explain the creation/redemption process of ETFs and the function of authorized participants
- 1.b.** Describe how ETFs are traded in secondary markets
- 1.c.** Describe sources of tracking error for ETFs
- 1.d.** Describe factors affecting ETF bid–ask spreads
- 1.e.** Describe sources of ETF premiums and discounts to NAV
- 1.f.** Describe costs of owning an ETF
- 1.g.** Describe types of ETF risk
- 1.h.** Identify and describe portfolio uses of ETFs

[LOS 1.a] Explain the creation/redemption process of ETFs and the function of authorized participants

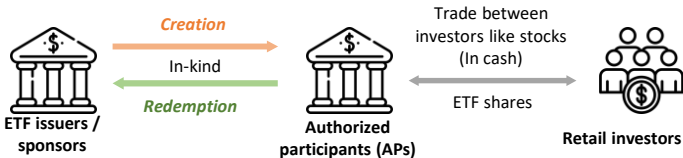
1.

Introduction to ETFs

Authorized participants

- Special group of institutional investors (large broker/dealers, often market makers).
- The only investors who can create or redeem new shares of an ETF (authorized by the ETF issuer).

How ETF shares are traded



Primary market (OTC)

Secondary market (Exchanges)

[LOS 1.a] *Explain* the creation/redemption process of ETFs and the function of authorized participants

1.

Introduction to ETFs

Differences from mutual funds

Mutual funds

Trade at the end of the day

Trade at closing NAV of the fund's holding

In cash (Cash-for share/share-for-cash swap)

Exchange-traded funds (ETFs)

Trade intraday

Trade at ETF prices (in line with NAVs but might be higher/lower)

In-kind (share-for-share swap)

[LOS 1.a] Explain the creation/redemption process of ETFs and the function of authorized participants

2.

Creation/redemption process

On primary market

Creation
process

ETF
issuers /
sponsors

Creation unit



Creation basket

Authorized
participants
(APs)

Redemption
process

ETF
issuers /
sponsors

Block of ETF shares that
AP no longer wants



Redemption basket

Authorized
participants
(APs)

- **Creation/Redemption basket** (list of required in-kind securities) is pre-specified and publicly disclosed daily by the ETF manager.
- ETF issuer specifies the size of a block of ETF shares as **creation unit** (usually 50,000 shares).

[LOS 1.a] Explain the creation/redemption process of ETFs and the function of authorized participants

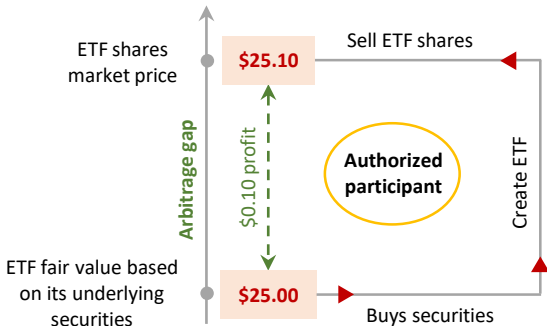
2.

Creation/redemption process

Incentive for APs to step in creation/redemption process is arbitrage.

ETF creation

Example:



Arbitrage gap

The price at which APs willing to step in and create or redeem shares

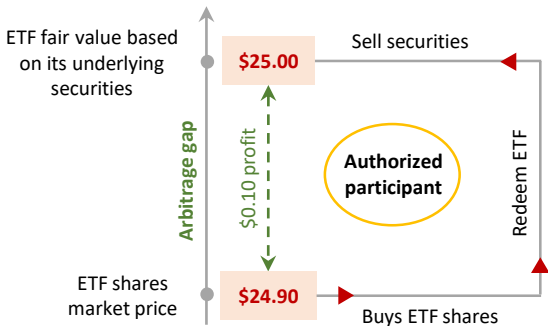
[LOS 1.a] Explain the creation/redemption process of ETFs and the function of authorized participants

2.

Creation/redemption process

ETF redemption

Example:



These profit-making scenarios do not include:

- The costs that the APs incur related to ETF trading or any fees the issuer may charge for creating or redeeming shares.
- Risk premium for illiquidity.

[LOS 1.a] *Explain* the creation/redemption process of ETFs and the function of authorized participants

2.

Creation/redemption process

Advantages of creation/redemption process

Lower cost

Tax efficiency

Keep market prices
in line with NAV

Lower cost

APs pass all costs in the ETF's bid–ask spread

→ Only affect transacting shareholders.

ETFs

Transacting shareholders sell/buy
fund shares

AP passed all costs in bid–ask spread

Transacting shareholders pay the
costs through bid–ask spread

Not affect non-transacting
shareholders

Mutual funds

Transacting shareholders sell/buy
fund shares

Fund sells/buys stocks

Incur additional costs for the fund

Affect non-transacting shareholders

[LOS 1.a] *Explain* the creation/redemption process of ETFs and the function of authorized participants

2.

Creation/redemption process

Advantages of creation/redemption process

Tax efficiency

Low-basis stocks

Stocks with low acquired costs → High unrealized gain

Because creation and redemption happen in kind

Create no realized gain/loss

Incur no tax

ETF issuers can choose low-basis stocks to redeem

Reduce the position's unrealized gain

Reduce tax

Keep market prices in line with NAV

APs will engage in arbitrage transactions if the ETFs trade at a price significantly different from their NAV.

→ Repeat until no further arbitrage opportunity exists.

→ Keep market prices in line with fair value.

[LOS 1.b] *Describe* how ETFs are traded in secondary markets

US Settlement



On exchange



Authorized
participants (APs)

2 days

Retail investors

National Security Clearing Corporation (NSCC) guarantees the performance of parties to a trade on an exchange

Depository Trust Company (DTC) transfers the securities

European Settlement



Over-the counter (OTC)



Authorized
participants (APs)

Retail investors

[LOS 1.c] Describe sources of tracking error for ETFs

Factors should be considered to best understand an ETF's ability to fit investor requirements are:

1**Expense ratios**

Actual costs to manage an
ETF

2**Index Tracking/Tracking
Error (Los 1.c)**

ETF's ability to track its
underlying index

3**Tax treatment**

ETFs have distinct tax
treatments that need to
evaluate

4**Potential costs and risks
(Los 1.d,e)**

ETFs have distinct costs and
risks that need to evaluate

[LOS 1.c] *Describe* sources of tracking error for ETFs

1.

Expense ratios

Expense ratio

Fund expense ratios measure the ***actual costs to manage an ETF***.

→ One of the first factors investors look at when evaluating ETFs.

ETFs

Not have to keep track of individual investor accounts

Not bear the costs of communicating directly with individual investors

Not require the security and macroeconomic research carried out by active managers

**Decrease
ETF's
expense
ratios**

Costs to manage an ETF depend on

Portfolio complexity

Issuer size

Competitive landscape

Number of securities held

Frequency of rebalancing or strategy implementation

Difficulty in maintaining portfolio exposures

[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

ETF managers attempt to tracks the fund performance as closely as possible.
→ Compare ETF performance with index returns using daily or periodic return.

Daily differences

Daily differences

Difference in returns between the fund, as measured by its NAV, and its index

Underlying
index's returns



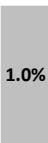
Daily difference

ETF's returns
measured by
NAV

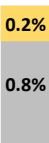


Example:

VN30's return
(31/1/2023)



1.0%



0.2%

0.8%

Daily difference

E1VFN30's
return
(31/1/2023)

Index tracking

Evaluated using the one-day difference in returns between the fund, as measured by its NAV, and its index

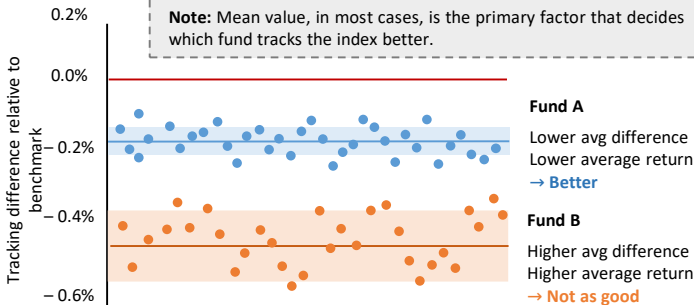
[LOS 1.c] *Describe* sources of tracking error for ETFs

Periodic tracking

The measure now has to include 2 factors: **(1) Central tendency** (mean, median of daily differences) **(2) variability** (standard deviation or range).

Tracking error

Standard deviation of daily differences, indicated the variability of a fund's excess return.



- Benchmark average return baseline
- Fund A daily difference
- Fund B average daily difference
- Fund A average daily difference
- Fund B tracking error
- Fund A tracking error

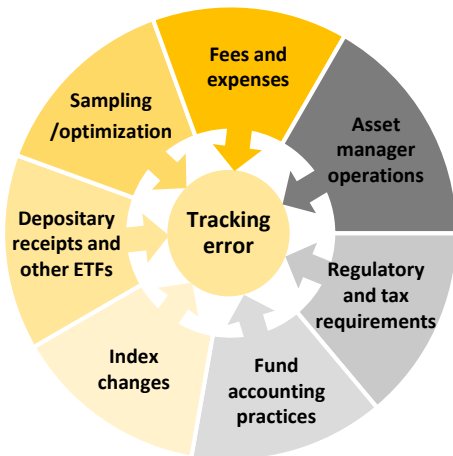
[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Sources of tracking error

Funds tracking the same underlying index can have very different index tracking results due to tracking error.



[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Fees and expenses

Fund operating expenses vary by ETF.

Index

Assume no costs
related to trading

ETFs

Have operating fees
and expenses

Reduce fund's return
relative to index

Example:

VN30's
return

Expense ratio

E1VFN30's
return

One would expect

ETF's
underperform
its benchmark

=

ETF's expense
ratio

on an annual basis

[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Representative sampling/optimization

For funds tracking small or illiquid markets index, owning every index constituent can be *difficult and costly*.

→ Funds may hold only a subset of index securities to track the benchmark index.

Make the median value
unpredictive

Dramatically expands the range of
results

S&P Asia

Pacific Emerging BMI Index

2,342
securities

Large-caps

Mid-caps

Small-caps

SPDR S&P Emerging
Asia Pacific ETF (GMF)

763 *large-caps*
securities

Mid-caps and small-caps

Outperform

large-caps

GMF underperform
the index

[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Depository receipts and other ETFs

Local market shares are illiquid

ETF managers choose to hold depository receipts (DRs) instead of local constituent shares

ETF issuers may choose to hold other ETFs as underlying holdings

Differences in trading hours

Differences in security prices

ETFs' holdings are valued at their closing market price (not their NAV)

Create discrepancies between fund NAV and index value

[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Index changes



Index provider

Occasionally rebalance or reorganize their indexes



ETF's portfolio managers

Delay in transact these changes at different prices

Create discrepancies in value

Fund accounting practices

Accounting practices of ETF and index are different (*such as time of day when value is calculated, exchange rate used,...*)

Magnify daily tracking differences

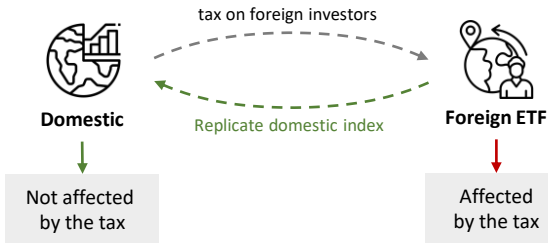
[LOS 1.c] *Describe* sources of tracking error for ETFs

2.

Index Tracking/Tracking Error

Regulatory and tax requirements

Regulatory and tax requirements may cause a fund to mis-track its index.



Asset manager operations

ETF issuers may:

- Lend a portion of their portfolio holdings to short sellers → Income for the portfolio.
- Work with foreign governments to minimize tax paid on distributions received (foreign dividend recapture).



Offset fund expenses

[LOS 1.c] *Describe* sources of tracking error for ETFs

3.

Tax issues

Tax-based evaluations for all ETFs

The likelihood of an ETF
distribution to shareholders

What happens when the investor
sells the ETF

**Capital gains
distributions**

**Other
distributions**

Taxes on sale

Capital Gains Distributions

Funds must distribute any capital gains realized during the year.

Advantages over traditional mutual funds

“Tax fair”

“Tax efficient”

[LOS 1.c] *Describe* sources of tracking error for ETFs

3.

Tax issues

Capital Gains Distributions

Tax fairness

The actions of investors selling shares of the fund do not influence the tax liabilities for remaining fund shareholders.

ETFs

Sell to another investors
in secondary market

Do not have to sell
portfolio to raise cash

No capital gain/loss

Incur no tax liability to
remaining shareholders

**Investors sell
shares back**

Mutual funds

Sell to another investors
in secondary market

Have to sell portfolio to
raise cash to pay investors

Have to sell portfolio to
raise cash to pay investors

Incur tax liability to
remaining shareholders

[LOS 1.c] *Describe* sources of tracking error for ETFs

3.

Tax issues

Capital Gains Distributions

Tax efficiency

The redemption process allows portfolio managers (ETF issuers) to manage the fund's tax liability.

By choosing shares with the largest unrealized capital gains (lowest cost basis)

- Use the in-kind redemption process to reduce potential capital gain
- Reduce tax

Other distributions

Other distributions
(dividend distribution)

Tax treatment varies by region, so investors must ensure they understand the tax treatment specific to each fund.

Taxes on sale

ETFs are taxed according to their underlying holdings when selling

Investor should consider tax treatment on different ETFs

[LOS 1.d] *Describe* factors affecting ETF bid–ask spreads

The bid–ask spread represents the market maker’s price for taking the costs and risk of the ETF transaction.

ETF Bid–Ask Spreads driver

Market structure

Opening hour and time zone in different markets

Different opening hour/time zone

Same opening hour/time zone

Underlying securities market open:

09.00am – 3.00pm

ETF market open:

10.00am – 4.00pm

Underlying securities market open:

09.00am – 3.00pm

ETF market open:

09.00am – 3.00pm

Wider bid–ask spreads

Liquidity of the underlying securities

Types of underlying held

Fixed-income securities

Large-cap stocks

Less liquid

More liquid

Wider bid–ask spreads

[LOS 1.d] *Describe* factors affecting ETF bid–ask spreads

ETF bid–ask spreads are generally less than or equal to the combination of the following:



Creation/redemption fees plus other trading costs



Spread of the underlying securities



Risk premium for carrying the trade until close of trading



AP's normal profit margin



Discount based on probability of offsetting the trade in secondary market

Do not contribute heavily in the spreads for very liquid ETFs

Maximum spread

- The quoted ETF bid–ask spread is generally for small trade size
- Larger trades: based on **liquidity conditions** and **volatility** in the market
→ Best handled via negotiation

[LOS 1.e] *Describe* sources of ETF premiums and discounts to NAV

1. >> ETF premiums and discounts

Premiums/discounts represents the different between ETFs market price and its NAV.

ETFs and underlying securities trade on the same exchange

Same market structure

Same closing price time

NAV is generally an accurate assessment of the ETF's fair value

Exchanges publish

Intraday price (ETF)

Market close

Closing price (ETF)



Indicated NAVs (iNAVs)

Intraday ETF "fair value" based on its creation basket

Intraday ETF premium or discount (%)

$$\frac{(\text{ETF price} - \text{iNAV per share})}{\text{iNAV per share}}$$

End-of-day NAV

Closing NAV that investor use to calculate returns

End-of-day ETF premium or discount (%)

$$\frac{(\text{ETF price} - \text{NAV per share})}{\text{NAV per share}}$$

[LOS 1.e] *Describe* sources of ETF premiums and discounts to NAV

2.

Sources of ETF premiums and discounts

Timing differences

NAV is often a poor fair value indicator for ETFs that

Hold foreign securities

OTC bonds

differences in exchange closing times

no true “closing prices” are available

NAV may be based on what those foreign securities would trade at if their local market was still open

ETF issuers rely on bids from bond desks or pricing services for proxy prices

Stale pricing

Stale price

Last ETF trade price of infrequently traded ETFs (have large premiums or discounts to NAV)

Last ETF trade:
2.00pm

ETF market and underlying market close:
4.00pm

NAV may differ from the **stale price** (2:00 pm price) of the ETF

[LOS 1.e] Describe sources of ETF premiums and discounts to NAV

2. Sources of ETF premiums and discounts



Illustration

Comparison of US ETF Trading Costs

ETF ticker	SPY	IVV	EUSA	IWM
Benchmark Index	S&P 500	S&P 500	MSCI USA Equal Weighted	Russell 2000

Volume in US dollars

Daily average volume	24.47 billion	1.22 billion	1.32 million	3.90 billion
Median volume	20.23 billion	1.08 billion	0.94 million	3.81 billion

Other trading characteristics

Average spread (%)	0.00%	0.01%	0.12%	0.01%
Average spread (\$)	\$0.01	\$0.03	\$0.07	\$0.01
Median premium/discount (%)	0.00%	0.00%	0.04%	0.01%
Maximum premium (%)	0.12%	0.13%	0.96%	0.12%
Maximum discount (%)	-0.19%	-0.11%	-0.38%	-0.13%

Assess characteristics of each ETF based on the above information.

[LOS 1.e] *Describe* sources of ETF premiums and discounts to NAV

2.

Sources of ETF premiums and discounts



Illustration

Comparison of US ETF Trading Costs

ETF ticker	SPY	IVV	EUSA	IWM
Benchmark Index	S&P 500	S&P 500	MSCI USA Equal Weighted	Russell 2000

Characteristics

<i>Liquidity</i>	_____	_____	_____	_____
<i>Spread</i>	_____	_____	_____	_____
<i>Premium/ discount</i>	_____	_____	_____	_____

[LOS 1.e] Describe sources of ETF premiums and discounts to NAV

2. Sources of ETF premiums and discounts



Illustration

Comparison of US ETF Trading Costs

ETF ticker	SPY	IVV	EUSA	IWM
Benchmark Index	S&P 500	S&P 500	MSCI USA Equal Weighted	Russell 2000

Characteristics

Liquidity (based on daily volume)	Most liquid	Less liquid	Low liquid	High liquid
Spread (based on average spread)	Tight spread	Tight spread	Wider spread	Tight spread
Premium/Discount (based on median, maximum premium/discount)	Tight premium/discount	Tight premium/discount	High premium/discount	Tight premium/discount

[LOS 1.f] *Describe* costs of owning an ETF

Total costs of ETF ownership

Cost of entering and exiting ETF positions

Commissions

Bid-ask spreads

Premiums/discounts
to NAV

Significant consideration for active short-term ETF investors

Other costs of ETF ownership

Management fees

Tracking error

Other costs

Significant for longer term ETF investors

[LOS 1.f] *Describe* costs of owning an ETF

Total costs of ETF ownership

Cost Factor Comparison—ETFs and Mutual Funds

Fund Cost Factor	Function of Holding Period?	Explicit /Implicit	ETFs	Mutual funds
Management fee	Y	Explicit	Often less	Yes
Tracking error	Y	Implicit	Often less	Index fund only
Commissions	N	Explicit	Some free	No
Bid–ask spread	N	Implicit	Yes	No
Premium/discount to NAV	N	Implicit	Yes	No
Portfolio turnover (from investor flows and fund management)	Y	Implicit	Often less	Yes
Taxable gains/losses to investors	Y	Explicit	Often less	Yes
Security lending	Y	Implicit	Often more	Yes

[LOS 1.f] *Describe* costs of owning an ETF

Total costs of ETF ownership

Trading costs vs. management fees

Example:



Investor



Amount: \$20,000 trade
3-month holding period
Pay: \$10 commission (0.05% each way)
0.15% bid–ask spread on purchase and sale
0.15% annual management fee

Holding period cost (%)

Round-trip trading cost (%)



Management fee for period (%)

One-way
commission
% × 2



$\frac{1}{2}$ Bid–ask
spread % × 2

Months hold / 12
(on an annual basis)

$$= (0.05\% \times 2) + (\frac{1}{2} \times 0.15\% \times 2)$$

$$= 3/12 \times 0.15\%$$

[LOS 1.f] *Describe* costs of owning an ETF

Total costs of ETF ownership

Trading costs vs. management fees



Illustration

Compare ETF management fee and trading cost in different holding period

Calculate and fill in the missing gaps using a 0.15% annual management fee

Holding Period	3 Months	12 Months	3 Years
Commission	0.10%	0.10%	0.10%
Bid–ask spread	0.15%	0.15%	0.15%
Management fee			
Total			
Trading costs (% of total)			
Management fees (% of total)			

[LOS 1.f] *Describe* costs of owning an ETF

Total costs of ETF ownership

Trading costs vs. management fees



Illustration

Compare ETF management fee and trading cost in different holding period

Holding Period	3 Months	12 Months	3 Years
Commission	0.10%	0.10%	0.10%
Bid–ask spread	0.15%	0.15%	0.15%
Management fee	0.0375%	0.15%	0.45%
Total	0.29%	0.40%	0.70%
Trading costs (% of total)	0.86%	0.625%	0.36%
Management fees (% of total)	0.14%	0.375%	0.64%

- **Short-term investors** should consider **trading costs** due to greater proportion of total costs.
- **Long-term investors** should consider **management fees** due to greater proportion of total costs.

[LOS 1.g] Describe types of ETF risk

Overview of ETF risk

Counterparty risk

Settlement risk

Security lending

Fund closures

Regulations

Competition

Corporate actions

*Creation and
redemption halts*

*Change in investment
strategy*

Investor-related risk

Leveraged ETFs risk

Inverse ETFs risk

[LOS 1.g] Describe types of ETF risk

1.

Counterparty risks

Some ETF legal structures expose the investors to counterparty risk.

ETNs

Unsecured debt obligations of the issuer (bank) that have high counterparty risk.

Example:

- Want to issue unsecured debt at fixed rate (10 %)
- Market unsecured debt fixed rate = 12%



Enter into an equity swap

Issue ETNs pay equity return

Pay SFR at 10%

Receive equity return

Pay equity return

Net off

Bank's effective borrowing costs = 10%
Bank does not hold underlying securities

Counterparty risk

[LOS 1.g] *Describe* types of ETF risk

1.

Counterparty risks

Settlement risk

Fund use OTC derivatives to gain market exposure

Mark-to-market gains subject to counterparty default

Mitigate by frequent settlement

Securities lending

Lend their underlying securities to short sellers to earn additional income

Risk of the security borrower defaulting is borne by ETF investors

Securities lent are generally overcollateralized

[LOS 1.g] *Describe types of ETF risk*

2.

Fund closures

ETF issuers may decide to close an ETF.

Reason for fund closures

Actual closures

Actual closures involve an actual fund closing.

Regulations

Security regulators can change the regulations governing certain types of funds.

Forced closure of those funds

Competition

High competition

Funds with low AUM and trading volume could be closed

Corporate actions

When ETF families merge or are sold to other ETF providers

New ETF owners may close underperforming ETFs

[LOS 1.g] *Describe types of ETF risk*

2.

Fund closures

ETF issuers may decide to close an ETF.

Reason for fund closures

“Soft” closures

“Soft” closures do not involve an actual fund closing.

Creation and redemption halts

Particular to ETNs, this risk arises when the issuer is no longer interested in additional borrowings.

→ ETNs may trade at a significant premium to their NAV as the arbitrage mechanism breaks down.

Change in investment strategy

Issuer may want to change investment strategy.

Close the fund tracking the original strategy and opening of a new fund tracking a different strategy

[LOS 1.g] *Describe* types of ETF risk

3.

Investor-related risk

ETFs provide access to sometimes complex asset classes and strategies.

→ May introduce risks to investors who do not fully understand them.

Leveraged and inverse funds generally offer levered and/or inverse exposure to a given index.

→ Must reset or adjust their exposure daily to deliver the target return multiple each day.



Illustration

Example of Levered 3× ETF Exposure

Consider a fund offering 300% exposure (3 times, or 3×) to the FTSE 100 Index with a net asset value of £100. It uses swaps to obtain a notional exposure of £300.

[LOS 1.g] Describe types of ETF risk

3. Investor-related risk



Illustration

Example of Levered 3× ETF Exposure

If the one-day FTSE 100 Index return is 5%, the £300 in exposure becomes £315 (a 5% increase).

→ ETF's end-of-day NAV is: $100 \times (1 + 3 \times 5\%) = £115$

In order to deliver 300% of the index's daily performance for the following day, the ETF, now valued at £115, requires notional exposure of £345 for 3 times exposure.

ETF has only £315 in exposure → must increasing notional swap exposure by £30.

	Index Level	One-Day Index Return (%)	3× ETF NAV (£)	Notional Swap Exposure (£)	3× Swap Exposure (£)	Swap Exposure Adjustment (£)
Day 1	100	—	100	300	300	0
Day 2	105	5%	115	315	345	30

[LOS 1.g] *Describe* types of ETF risk

3.

Investor-related risk



Illustration

Example of Levered and Inverse 2× ETF Daily Return vs. Holding Period Return

Consider a fund offering 2 times (−200%) exposure to the S&P 500 Index. The fund (−2× ETF) has a starting net asset value of \$100 and uses swaps to obtain notional exposure.

Day 1: Both the index and the −2× fund are at a starting level of 100.

[LOS 1.g] Describe types of ETF risk

3. Investor-related risk



Illustration

Example of Levered and Inverse 2× ETF Daily Return vs. Holding Period Return

Day 2: The index increases to 110, a one-day return of 10%.

The −2× ETF daily return is calculated as follows:

$$= -2 \times [(110 - 100)/100] = -2 \times (10\%) = -20\%.$$

The −2× ETF NAV is calculated as follows:

$$= 100 \times (1 + -0.2) = 80.$$

	Index Level	One-Day Index Return (%)	Index Period Return (%)	−2× ETF NAV	One Day ETF Return (%)	2× ETF Holding Period Return (%)
Day 1	100	—		100	—	—
Day 2	110	10%	10%	80	−20%	−20%

[LOS 1.g] Describe types of ETF risk

3.

Investor-related risk



Illustration

Example of Levered and Inverse 2× ETF Daily Return vs. Holding Period Return

Day 3: The index falls to 99, a one-day return of −10%.

The 2× ETF daily return is calculated as follows:

$$= -2 \times [(99 - 110)/110] = -2 \times (-10\%) = 20\%.$$

The −2× ETF NAV is calculated as follows:

$$= 80 \times (1 + 0.2) = 96.$$

	Index Level	One-Day Index Return (%)	Index Period Return (%)	−2× ETF NAV	One Day ETF Return (%)	2× ETF Holding Period Return (%)
Day 1	100	—		100	—	—
Day 2	110	10%	10%	80	−20%	−20%
Day 3	99	−10%	−1%	96	20%	−4%

[LOS 1.g] Describe types of ETF risk

3.

Investor-related risk



Illustration

Example of Levered and Inverse 2× ETF Daily Return vs. Holding Period Return

Over the three days, the index return is -1% : $(99 - 100)/100$. A naive expectation might assume that over the same period, the $-2\times$ ETF would return 2% ($= -2 \times -1\%$).

Over the three days, the fund's actual return was -4% : $(96 - 100)/100$.

	Index Level	One-Day Index Return (%)	Index Period Return (%)	$-2\times$ ETF NAV	One Day ETF Return (%)	$2\times$ ETF Holding Period Return (%)
Day 1	100	—		100	—	—
Day 2	110	10%	10%	80	-20%	-20%
Day 3	99	-10%	-1%	96	20%	-4%

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

Application of ETFs

Portfolio efficiency (1.)

The use of ETFs to better manage a portfolio for efficiency or operational purposes

Asset class exposure management (2.)

The use of ETFs to achieve or maintain core exposure or investment themes

Active and factor investing (3.)

The use of ETFs to target specific active or factor exposures

1.

Efficient portfolio management

Portfolio liquidity management

Portfolio

Benchmark
to VN30

Excess
cash

Equitize excess cash

Invest in ETFs
(E1VFN30)

Minimize cash drag

Lower trading costs and
easier operationally

Cash drag: fund's mis-tracking relative to its index that results from holding uninvested cash

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

Portfolio rebalancing

- ETFs can be used to cost-effectively rebalance portfolios to target specific asset class weights.
- For investors who have the ability to sell short, ETFs can also be shorted to quickly reduce the weight of a specific sector or asset class.

Portfolio completion strategies

ETFs can also be used to fill a temporary gap in exposure to an asset class, sector, or investment theme or factor.

Gaps arise when

New manager has different macro views

Existing manager's allocation differs from the investor's desired exposure

A manager moves out of small-caps

Investor seeking continued exposure to small-caps



invest a portion of her portfolio in a small-cap ETF

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

Transition management

Transition management

Hiring and firing managers

Making changes to allocations with existing managers



New manager can

Selling the unwanted positions of the manager she is replacing

Invest in an ETF to maintain market exposure

Drawbacks of using ETFs with very large asset owners

Instead of invest directly to ETF, very large asset owners can invest through a separately managed account (SMA).

May be able to
negotiate lower fees
for a SMA

SMA can be
customized to the
investment goals and
needs of the investor

Regulators may
require large ETF
holdings to be
disclosed to the public

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

2.

Asset class exposure management

With a wide variety of asset class, subclass, and sector, ETFs provide investors with a more efficient and liquid means relative to investing in the underlying securities (especially fixed-income).

Core exposure to an asset class or sub-asset class

ETFs' diversity across global equities, bonds, commodities, and currencies

Primary strategic use of ETFs is to **gain core index exposure** to various asset classes and sub-asset classes **based on suitability** to investors.

Tactical strategies

- Implement market views and adjust portfolio risk on a more **short-term, tactical basis** → rotate money into/out of sectors expected to perform better/worse using ETFs.
- ETFs selected for short-term tactical strategies are selected based on lower trading cost and high liquidity rather than low management fees.

Thematic ETFs

Select subsectors that are expected to outperform.
E.g., ecommerce versus the broad technology sector

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

3.

Active and Factor Investing

Factor (smart beta) ETFs

Used to gain systematic active exposure to persistent common return drivers or factors

Represent approximately 20% of global ETF assets

Active ETFs

Investment strategy is benchmarked but managed with discretion

Represent approximately 2-3% of global ETF assets

Factor (smart beta) ETFs

Factor ETFs

- Are benchmarked to an index created with predefined rules for screening and/or weighting constituent holdings.
- The ETF strategy is based on return drivers.
- ETFs' managers may compete on the factor chosen or the weights assigned to portfolio holdings.



- Provide longer-term, buy-and-hold exposure to a desired factor based on an investment view.
- Add risk factor allocations that might not be present in a benchmark or portfolio.

[LOS 1.h] *Identify and describe portfolio uses of ETFs*

3.

Active and Factor Investing

Factor (smart beta) ETFs

Multi-factor ETFs

- Provide exposure to several factors and can dynamically change portfolio weights based on market opportunities.
- Strategy design involves factor selection, factor strategy construction, and a weighting scheme across factors that is managed over time.

The success of active strategy ETFs is related to

Whether the ETF was
successful in **gaining
exposure to that factor**

Whether the **chosen exposure
was rewarded** by the market

[LOS 1.h] *Identify and describe* portfolio uses of ETFs**3.****Active and Factor Investing****Risk management**

- Some ETFs are constructed to provide higher or lower risk relative to a passive index.
- Low volatility ETFs use rules to construct portfolios with low relative return volatility.
- Some ETFs may be constructed to hedge specific risks (currency-hedged ETFs, duration-risk-hedged ETFs).

Alternatively weighted ETFs

Constructed using portfolio weights that differ from standard market cap weight.

E.g., ETFs that weight constituent stocks on the basis of their dividend yields.
→ Select or overweight stocks with higher dividend yields
→ Used by investors seeking income-generating strategies

[LOS 1.h] *Identify and describe* portfolio uses of ETFs**3.****Active and Factor Investing****Discretionary active ETFs**

- Actively managed and are similar to closed-end mutual funds.
- The largest of these are fixed-income ETFs, which include exposures to senior bank loans, mortgage securities, and floating rate notes.

Dynamic asset allocation and multi-asset strategies

- Dynamic top-down asset allocation ETFs that invest in stocks and bonds based on risk/return forecasts.
- Popular among global asset managers and hedge funds for their discretionary asset allocation.



MODULE 2: USING MULTIFACTOR MODELS



LEARNING OUTCOMES

- 2.a.** Describe arbitrage pricing theory (APT), including its underlying assumptions and its relation to multifactor models.
- 2.b.** Define arbitrage opportunity and determine whether an arbitrage opportunity exists.
- 2.c.** Calculate the expected return on an asset given an asset's factor sensitivities and the factor risk premiums.
- 2.d.** Describe and compare macroeconomic factor models, fundamental factor models, and statistical factor models.
- 2.e.** Describe uses of multifactor models and interpret the output of analyses based on multifactor models.
- 2.f.** Describe the potential benefits for investors in considering multiple risk dimensions when modeling asset returns.
- 2.g.** Explain sources of active risk and interpret tracking risk and the information ratio.

Overview

**Arbitrage pricing
theory (APT)**



Multifactor models

***Macroeconomic
factor models***

***Fundamental
factor models***

***Statistical factor
models***

[LOS 2.a] *Describe* arbitrage pricing theory (APT), including its underlying assumptions and its relation to multifactor models.

1.

Assumptions of Arbitrage Pricing Theory (APT)

Arbitrage pricing theory (APT)

A linear model with multiple systematic risk factors *priced* by the market. However, APT does not identify the specific risk factors.

Assumptions

APT makes less strong assumptions compared to CAPM.

Unsystematic risk can be diversified in a portfolio

Investors have the choice of a large number of assets
→ unsystematic risk can be diversified by forming portfolios

Returns are generated using a factor model

APT provides little practical guidance for identifying the risk factors → Major weakness of APT.

No arbitrage opportunities exist

Investors will undertake infinitely large positions to exploit any perceived mispricing → asset prices to adjust immediately to their equilibrium values

[LOS 2.a] *Describe* arbitrage pricing theory (APT), including its underlying assumptions and its relation to multifactor models.

2.

The APT Equation

The APT describes the equilibrium relationship between expected returns for welldiversified portfolios and their multiple sources of systematic risk.

$$E(RP) = R_F + \beta_{p,1}(\lambda_1) + \beta_{p,2}(\lambda_2) + \dots + \beta_{p,k}(\lambda_k)$$

- λ_j : risk premium for a portfolio (called a ***pure factor portfolio***) with factor sensitivity equal to 1 to factor j and factor sensitivities of zero for the remaining factors
- β : factor sensitivity of Portfolio P to that risk factor.

Unlike the CAPM, the APT does not require that one of the risk factors is the market portfolio.

→ Major advantage of the arbitrage pricing model

[LOS 2.b] *Define* arbitrage opportunity and *determine* whether an arbitrage opportunity exists.

1.

Method for exploiting arbitrage opportunities

The APT assumes there are **no market imperfections** preventing investors from exploiting arbitrage opportunities.

Extreme long and short positions are permitted and **mispricing will disappear** immediately.

All arbitrage opportunities would be **exploited and eliminated** immediately.



Example

Example: Exploiting an arbitrage opportunity.

Suppose your investment firm uses a single-factor model to evaluate assets. Consider the following data for Portfolios A, B, and C:

Portfolio	Expected return	Factor sensitivity (Beta)
A	10%	1.0
B	20%	2.0
C	30%	1.5

Calculate the arbitrage opportunity from the data provided.

[LOS 2.b] *Define* arbitrage opportunity and *determine* whether an arbitrage opportunity exists.

1.

Method for exploiting arbitrage opportunities



Example

Answer:

Step 1: Construct a new portfolio from two given portfolios which has the same level of systematic risk with the remaining portfolio.

By allocating 50% of our funds to Portfolio A and 50% to Portfolio B, we can obtain a Portfolio (D) with beta *equal* to the Portfolio C beta (1.5):

$$\text{Beta for Portfolio D} = 0.5(1) + 0.5(2) = 1.5$$

While the betas for Portfolios D and C are identical, the expected returns are different:

$$\text{Expected return for Portfolio D} = 0.5(0.10) + 0.5(0.20) = 0.15 = 15\%$$

Step 2: Enter arbitrage positions by buying over-performing portfolio and selling under-performing portfolio.

From step 1, we have created Portfolio D that has the same risk as Portfolio C (beta = 1.5) but has a higher expected return than Portfolio C.

→ By purchasing Portfolio D and short-selling Portfolio C, we expect to earn a 2% return (15% minus 13%).

[LOS 2.b] *Define* arbitrage opportunity and *determine* whether an arbitrage opportunity exists.

1.
Method for exploiting arbitrage opportunities

Example

Answer (con't):

- The portfolio that is long Portfolio D and short Portfolio C is called the ***arbitrage portfolio***.
- We have invested nothing upfront because we can use the proceeds of the short sale on Portfolio C to purchase Portfolio D.
- The overall beta of our investment equals the difference in betas between our long and short positions: $1.5 - 1.5 = 0$; we have undertaken no net systematic risk.
- As investors exploit the arbitrage opportunity, prices of assets in Portfolio C will drop and the (future) expected return for Portfolio C will rise to its equilibrium value.

[LOS 2.c] Calculate the expected return on an asset given an asset's factor sensitivities and the factor risk premiums.

Given a portfolio's factor exposures (betas) and factor risk premiums, we can easily compute the portfolio's expected return

 Example

Example: Calculating expected returns from the arbitrage pricing model

An investment firm employs a two-factor APT model. The risk-free rate equals 5%. Determine the expected return for the Invest Fund using the following data:

	Factor 1	Factor 2
Invest Fund factor betas	1.50	2.00
Factor risk premiums	0.0300	0.0125

Answer:

Using the two-factor APT model, the expected return for the Invest Fund (IF) equals:

$$E(R_{IF}) = 0.05 + 1.5(0.03) + 2(0.0125) = 0.12 = 12\%$$

[LOS 2.c] Calculate the expected return on an asset given an asset's factor sensitivities and the factor risk premiums.

We can also use factor models to compute the parameter values given expected returns and factor exposures.

 Example

Example: Calculating APT parameters given expected returns

1. Given a one-factor model and the following information, calculate the risk-free rate and the factor risk premium.

Portfolio	Expected return	Factor sensitivity
A	7.0%	1.0
B	7.8%	0.0125

2. Verify that Portfolio C with an expected return of 6.2% and factor sensitivity of 0.8 is priced correctly.

[LOS 2.c] Calculate the expected return on an asset given an asset's factor sensitivities and the factor risk premiums.



Example

Answer:

1. As for one-factor model, expected return is:

Expected return = risk-free rate + factor sensitivity $\times$ risk premium

Therefore, given the information for portfolio A:

$$0.07 = R_f + 1.0 \times \lambda$$

$$R_f = 0.07 - \lambda$$

Substituting this information for Portfolio B:

$$0.078 = (0.07 - \lambda) + 1.2\lambda;$$

$$\lambda = 0.04 \text{ or } 4\%$$

$$R_f = 0.07 - \lambda = 0.07 - 0.04 = 0.03 \text{ or } 3\%$$

2. Expected return for Portfolio C = $0.03 + (0.8 \times 0.04) = 6.2\%$. Hence, Portfolio C is correctly priced.

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

A **multifactor model** assumes asset returns are driven by more than one factor. Has three classifications:

**Macroeconomic
factor models**

Assume that asset returns are explained by **surprises** in macroeconomic risk factors.

Factor surprises: the difference between the realized value of the factor and its consensus predicted value.

**Fundamental
factor models**

Assume asset returns are explained by multiple firmspecific factors (e.g., P/E ratio, market cap, leverage ratio, and earnings growth rate)

**Statistical factor
models**

Use statistical methods to explain asset returns. Two types:

- **Factor analysis:** factors are portfolios that explain **covariance** in asset returns.
- **Principal component models:** factors are portfolios that explain the **variance** in asset returns.

Weakness: factors do not lend themselves well to economic interpretation. → statistical factors are **mystery factors**.

[LOS 2.d] Describe and compare macroeconomic factor models, fundamental factor models, and statistical factor models.

1.

Macroeconomic factor models

The following model is an example of a two-factor macroeconomic model in which stock returns are explained by surprises in **GDP growth rates** and **credit quality spreads**:

$$R_i = E(R_i) + b_{i1}F_{GDP} + b_{i2}F_{QS} + \epsilon_i$$

R_i = return for Asset i

$E(R_i)$ = expected return for Asset i (in the absence of any surprise)

F_{GDP} = surprise in the GDP rate

F_{QS} = surprise in the credit quality spread (BB-rated bond yield – Treasury bond yield)

b_{i1} = GDP surprise sensitivity of Asset i

b_{i2} = credit quality spread surprise sensitivity of Asset i

ϵ_i = firm-specific surprise (unrelated to the two macro factors)

Factor surprise (F)

The difference between the predicted value of the factor and the realized value

Sensitivity to each surprise (b)

The higher the sensitivity, the larger the change in return for a given factor surprise

Firm-specific surprise

Part of the return that can't be explained by the model, representing unsystematic risk.

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

1.

Macroeconomic factor models

The main features of the macroeconomic factor model include the ***systematic*** or ***priced risk factors*** and the ***factor sensitivities***.

Priced Risk Factors (F)

- ***Systematic risk*** factors will affect even well-diversified portfolios, representing **priced risk**.
- investors can expect reward.

Factor Sensitivities (b)

- The ***factor sensitivities*** of the model can be estimated by regressing historical asset returns on the corresponding historical macroeconomic factors.

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

1.

Macroeconomic factor models



Example

Example: Compute a stock return using a macroeconomic factor model

The following two-factor model is used to explain the returns for Media Tech (MT):

$$R_{MT} = E(R_{MT}) + b_{MT,1}F_{GDP} + b_{MT,2}F_{QS} + \varepsilon_{MT}$$

The expected return for Media Tech equals 10%. Over the past year, GDP grew at a rate that was 2 percentage points higher than originally expected, and the quality spread was 1 percentage point lower than originally expected. Media Tech's sensitivity to the GDP rate factor equaled 2, and its sensitivity to the quality spread factor equaled -0.5 . Over the past year, Media Tech also experienced a 2% company-unique surprise return (i.e., unrelated to the two macro factors). Construct the macroeconomic factor model for Media Tech, and calculate its return for the year.

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

1.

Macroeconomic factor models



Example

Answer:

Step 1: Construct multi-factor model

The two-factor model for Media Tech return is:

$$R_{MT} = E(R_{MT}) + b_{MT,1}F_{GDP} + b_{MT,2}F_{QS} + \epsilon_{MT}$$

Step 2: Use information about factor sensitivity and factor shock into the model:

$$R_{MT} = 0.10 + 2(0.02) - 0.5(-0.01) + 0.02 = 0.165, \text{ or } 16.5\%$$

In which:

The expected return: 0.100

Return from the positive GDP surprise: 0.040

Return from the positive quality spread surprise: 0.005

Return from unexpected firm specific event: 0.02

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

1.

Macroeconomic factor models



Example

Answer (con't):

The Media Tech return was higher than originally expected because MT was positively affected by:

- Higher-than-expected economic growth (GDP)
- Lower-than expected credit quality risk spreads (QS)
- Positive company-specific surprise events.

Be careful to interpret the signs properly. A decrease in the **quality spread** (a surprise less than zero) is good news for MT stock because it has a negative sensitivity to the factor. When credit quality spreads increase, MT's return goes down, and when credit quality spreads decrease, MT's return goes up.

[LOS 2.d] Describe and compare macroeconomic factor models, fundamental factor models, and statistical factor models.

2.

Fundamental factor models

Consider the following fundamental factor model:

$$R_i = a_i + b_{i1}F_{P/E} + b_{i2}F_{SIZE} + \epsilon_i$$

R_i = return for stock i

$F_{P/E}$ = return associated with the P/E factor

F_{SIZE} = return associated with the SIZE (market capitalization) factor

a_i = intercept

b_{i1} = standardized sensitivity of stock i to the P/E factor

b_{i2} = standardized sensitivity of stock i to the SIZE factor

ϵ_i = portion of asset i return not explained by the factor model

Standardized sensitivities
(b_{i1} and b_{i2})

Sensitivities in most fundamental factor models are **NOT** regression slopes, **BUT** are standardized attributes (similar to z-statistics). (Please refer to next slide for formula)

Factor returns
($F_{P/E}$ and F_{SIZE})

Slopes of cross-sectional regressions in which the dependent variable is the set of returns for all stocks and the independent variables are the standardized sensitivities.

Intercept terms
(a_i)

In fundamental factor models, the factors are not return surprises. Hence, the intercept term is no longer interpreted as the expected return.

[LOS 2.d] Describe and compare macroeconomic factor models, fundamental factor models, and statistical factor models.

2.

Fundamental factor models

For example, the standardized P/E sensitivity in a fundamental factor model is calculated as:

$$b_{i1} = \frac{P/E_i - \overline{P/E}}{\sigma_{P/E}}$$

Where:

P/E_i : P/E for stock i

$\overline{P/E}$: average P/E calculated across all stocks

$\sigma_{P/E}$: standard deviation of P/E ratios across all stocks

By standardizing the factor sensitivity, we measure the number of standard deviations that each sensitivity is from the average.

It allows us to use fundamental factors measured in different units in the same factor model.

For example: a stock with a standardized P/E sensitivity of 2.0 has a P/E that is 2 standard deviations above the mean.

For example: P/E ratios are usually greater than 1.00, while dividend yields are in percentages (i.e., less than 1.00).

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

2.
Fundamental factor models

Example

Example: Calculating a standardized sensitivity in a fundamental factor model.

The P/E for stock i is 15.20, the average P/E for all stocks is 11.90, and the standard deviation of P/E ratios is 6.30. Calculate the standardized sensitivity of stock i to the P/E factor.

Answer:

The sensitivity of stock i to the P/E factor is:

$$\beta_{i,P/E} = \frac{15.20 - 11.90}{6.30} = 0.52$$

Therefore, the P/E ratio for the stock is 0.52 standard deviations higher than the average stock P/E.

[LOS 2.d] *Describe and compare* macroeconomic factor models, fundamental factor models, and statistical factor models.

2.

Comparison of two model types

Key differences between the macroeconomic factor model and the fundamental factor model

	Fundamental factor model	Macroeconomic factor model
Sensitivities	Are calculated directly from the attribute (e.g., P/E) data and not estimated.	Are regression slope estimates.
Factors interpretation	Are rates of return associated with each factor and are estimated using multiple regression.	Are surprises in the macroeconomic variables.
Intercept terms	Has no economic interpretation and is simply necessary to make the unsystematic risk of the asset equal to zero.	Equals the stock's expected return from an equilibrium pricing model like the APT.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

Multifactor models can be useful for *return attribution*, *risk attribution*, and *portfolio construction*.

1.

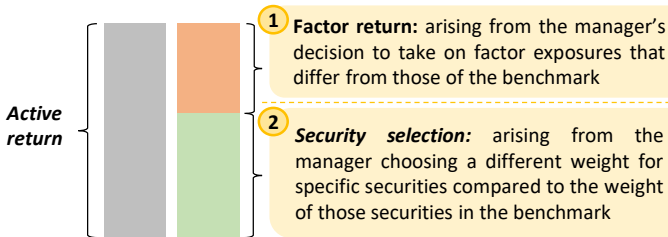
Return attribution

Active return

Equals the differences in returns between a managed portfolio and its benchmark:

$$\text{Active return} = R_p - R_b$$

Components of active return



[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

Multifactor models can be useful for *return attribution*, *risk attribution*, and *portfolio construction*.

1.

Return attribution

Components of active return

Active return

Active return = factor return + security selection return

Where:

β_{pi} = factor sensitivity for the i th factor in the active portfolio

β_{bi} = factor sensitivity for the i th factor in the benchmark portfolio

λ_i = factor risk premium for factor i

1 Factor return

$$\text{Factor return} = \sum_{i=1}^k (\beta_{pi} - \beta_{bi}) \times \lambda_i$$

2 Security selection return

security selection return = active return – factor return

The security selection return is then the **residual difference** between active return and factor return

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

1.

Return attribution



Example

Example: Return decomposition

Glendale Pure Alpha Fund generated a return of 11.2% over the past 12 months, while the benchmark portfolio returned 11.8%. Suppose we are provided a fundamental factor model with two factors as given in the following.

Factor	Factor sensitivity (betas)		Factor risk premium (λ)
	Portfolio	Benchmark	
P/E	1.10	1.00	-5.00%
Size	0.69	1.02	2.00%

1. Attribute the cause of the difference in returns.
2. Describe the manager's apparent skill in factor bets as well as in security selection.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

1.

Return attribution

Answer:

Factor	Factor sensitivity (betas)			Factor risk premium	Contribution to Active return
	Portfolio	Benchmark	Difference		
	(1)	(2)	(3) = (1) – (2)	(4)	(5) = (3) x (4)
P/E	1.10	1.00	0.10	-5.00%	-0.50%
Size	0.69	1.02	-0.33	2.00%	-0.66%
Total					-1.16%

1. Active return is the difference between portfolio return and benchmark return
 $11.20\% - 11.80\% = -0.60\%$

Recall: Contribution to Active return of each Factor

$$= (\text{Portfolio return} - \text{Benchmark return})_{\text{factor}} \times \text{Factor risk premium}$$

And:

Factor tilts return + Security selection return = Active return

Contribution of P/E factor: $(1.10 - 1.00) \times -5.00\% = -0.50\%$

Contribution of Size factor: $(0.69 - 1.02) \times 2.00\% = -0.66\%$

Total contribution of factor tilts to active return: $-0.50\% - 0.66\% = -1.16\%$

Return from security selection = $-0.6\% - (-1.16\%) = +0.56\%$

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

1.

Return attribution

Answer (con't):

2. The active manager's regrettable factor bets resulted in a return of -1.16% relative to the benchmark. However, the manager's superior security selection return of $+0.56\%$ resulted in a total active return of -0.60% relative to the benchmark.

[LOS 2.e] Describe uses of multifactor models and interpret the output of analyses based on multifactor models.

2.

Risk Attribution

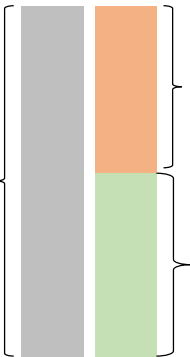
Active risk

Active risk (also known as *tracking error* or *tracking risk*) is defined as the standard deviation of the active return:

$$\text{Active risk} = \text{tracking error} = \sigma(R_P - R_B)$$

Components of active risk

Active risk squared



- 1 Active factor risk:** Risk from active factor tilts attributable to deviations of the portfolio's factor sensitivities from the benchmark's sensitivities to the same set of factors.

Example: A portfolio manager may decide to under- or overweight particular industries relative to the portfolio's benchmark.

- 2 Active specific risk:** Risk from active asset selection attributable to deviations of the portfolio's individual asset weightings versus the benchmark's individual asset weightings.

Example: The active portfolio manager may decide to overweight or underweight individual stocks within specific industries.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

2.

Risk Attribution

Active risk squared

Active risk squared = active factor risk + active specific risk

Where:

W_{pi} and W_{bi} = weight of i th security in the active and benchmark

$\sigma_{\epsilon_i}^2$ = residual (unsystematic) risk of the i th asset

1 *Active factor risk*

Active factor risk = active risk squared – active specific risk

Active factor risk represents the risk explained by deviation of the portfolio's factor exposures relative to the benchmark and is computed as the residual

2 *Active specific risk*

Active specific risk = $\sum_{i=1}^n (W_{pi} - W_{bi})^2 \times \sigma_{\epsilon_i}^2$

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

2.

Risk Attribution



Example

Example: Risk decomposition

Steve Martingale, CFA, is analyzing the performance of three actively managed mutual funds using a two-factor model. The results of his risk decomposition are shown in the following table:

Fund	Active factor			Active specific	Active risk squared
	Size factor	Style factor	Total factor		
Alpha	6.25	12.22	18.47	3.22	21.69
Beta	3.20	0.80	4.00	12.22	16.22
Gamma	17.85	0.11	17.96	19.7	37.66

1. Which fund assumes the highest level of active risk?
2. Which fund assumes the highest level of style factor risk as a proportion of active risk?
3. Which fund assumes the highest level of size factor risk as a proportion of active risk?
4. Which fund assumes the lowest level of active specific risk as a proportion of active risk?

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

2.

Risk Attribution



Example

Answer:

The following table shows the proportional contributions of various sources of active risk as a proportion of active risk squared. For example, the proportional contribution of style factor risk for Alpha fund can be calculated as $12.22 / 21.69 = 56\%$.

Fund	Active factor			Active specific	Active risk
	Size factor	Style factor	Total factor		
Alpha	29%	56%	85%	15%	4.7%
Beta	20%	5%	25%	75%	4.0%
Gamma	47%	0%	48%	52%	6.1%

1. The Gamma fund has the highest level of active risk (6.1%). Note that active risk is the square root of active risk squared (as given).
2. The Alpha fund has the highest exposure to style factor risk as seen by 56% of active risk being attributed to differences in style.
3. The Gamma fund has highest exposure to size factor as a proportion of total active risk (47%) compared to the other two funds.
4. The Alpha fund has the lowest exposure to active specific risk (15%) as a proportion of total active risk.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

3.

Uses of Multifactor Models

Multifactor models can be useful to a **passive** manager who seeks to replicate the factor exposures of a benchmark, or to an **active** manager who seeks to make directional bets on specific factors. Applications include:

Passive management

Managers seeking to track a benchmark can construct a ***tracking portfolio*** which has a deliberately designed set of factor exposures to match (track) a predetermined benchmark.

Active management

Active managers use factor models to make specific bets on desired factors while hedging (or remaining neutral) on other factors. ***Factor portfolios*** are particularly useful for speculation or hedging purposes.

For example: suppose that a portfolio manager believes GDP growth will be stronger than expected but wishes to hedge against all other factor risks. The manager can take a long position in the GDP factor portfolio.

Rules-based or algorithmic active management (alternative indices).

These strategies use rules to mechanically tilt factor exposures when constructing portfolios. These strategies introduce biases in the portfolio relative to valuetype benchmark indices.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

4.

Carhart Model

The **Carhart four-factor model** is a multifactor model that extends the Fama and French three-factor model to include not only market risk, size, and value as relevant factors, but also momentum.

Carhart model

$$E(R) = R_F + \beta_1 RMRF + \beta_2 SMB + \beta_3 HML + \beta_4 WML$$

Where:

$E(R)$ = expected return

R_F = risk-free rate of return

RMRF = return on value-weighted equity index – the risk-free rate

SMB = average return on small cap stocks – average return on large cap stocks

HML = average return on high book-to-market stocks – average return on low book-to-market stocks

WML = average return on past winners – average return on past losers

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

4.

Carhart Model



Example

Example: Factor portfolios

Sam Porter is evaluating three portfolios based on the Carhart model. The following table provides the factor exposures of each of these portfolios to the four Carhart factors:

Portfolio	Risk factors			
	RMRF	SMB	HML	WML
Eridanus	1.0	0.0	0.0	0.0
Scorpius	0.0	1.0	0.0	0.0
Lyra	1.2	0.0	0.2	0.8

Which strategy would be most appropriate if the manager expects that:

1. RMRF will be higher than expected.
2. Large cap stocks will outperform small cap stocks.

[LOS 2.e] *Describe* uses of multifactor models and *interpret* the output of analyses based on multifactor models.

4.
Carhart Model

Example
Answer:
1. RMRF will be higher than expected:

- The manager would go long in the Eridanus portfolio as it is constructed to have exposure only to the RMRF factor.
- The Lyra portfolio would not be ideal for Porter's purpose because it provides unneeded exposures to the HML and WML factors as well.

2. Large cap stocks will outperform small cap stocks:

- The manager would go short the Scorpius portfolio, which is constructed to be a pure bet on SMB (i.e., Scorpius is a factor portfolio).
- We short the portfolio because we are expecting that large cap stocks will outperform small cap stocks.

[LOS 2.f] *Describe* the potential benefits for investors in considering multiple risk dimensions when modeling asset returns.

CAPM framework

Investors choose a combination of the market portfolio and the risk-free asset depending on their risk tolerance.

Multifactor models

Investors can zero in on risks that the investor has a comparative advantage in bearing and avoid the risks that the investor is incapable of absorbing.

Example:

A pension plan invests for long-term and, hence, would not be averse to holding a security that bears liquidity risk (and that offers a liquidity risk premium).

[LOS 2.g] Explain sources of active risk and interpret tracking risk and the information ratio.

1.

The information ratio

Active return alone is insufficient for measuring an investment manager's performance over a series of measurement periods.

Example:

Imagine that Manager A earned a constant 0.5% (50 bps) active return over each of the last four quarters. Furthermore, suppose Manager B earned active returns of 8%, 5%, -3%, and -8% over the same four quarters. The average active returns for managers A and B are both 0.5%, but Manager B experienced far more volatility (i.e., less consistency) than Manager A.

To demonstrate a manager's consistency in generating active return, we use the **information ratio**, which standardizes average active return by dividing it by its standard deviation.

Information ratio

$$IR = \frac{\overline{R_P} - \overline{R_B}}{\sigma(R_P - R_B)}$$

[LOS 2.g] Explain sources of active risk and interpret tracking risk and the information ratio.

1.

The information ratio

Example: Calculating the information ratio

Imagine that the portfolio and benchmark returns over the past 12 months have been as shown in the following table.

Portfolio and Benchmark Returns for Twelve Months

Month	R_p	R_B	$R_p - R_B$
1	0.0101	0.0091	0.0010
2	-0.0013	0.0062	-0.0075
...			
11	0.0186	0.0012	0.0174
12	0.0103	0.0097	0.0006
Average	0.0076	0.0059	0.0017
Sample Std. Dev.			0.0063

Given the data in the table, calculate and interpret the manager's information ratio.

[LOS 2.g] Explain sources of active risk and interpret tracking risk and the information ratio.

1.

The information ratio

Answer:

$$IR = \frac{\overline{R_P} - \overline{R_B}}{\sigma(R_P - R_B)} = \frac{0.0076 - 0.0059}{0.0063} \approx 0.27$$

The higher the IR, the more active return the manager earned per unit of active risk. An information ratio of 0.27 indicates the manager earned about 27 basis points of active return per 1 percent of active risk.



MODULE 3: MEASURING AND MANAGING MARKET RISK



LEARNING OUTCOMES

- 3.a.** Explain the use of value at risk (VaR) in measuring portfolio risk.
- 3.b.** Compare the parametric (variance–covariance), historical simulation, and Monte Carlo simulation methods for estimating VaR.
- 3.c.** Estimate and interpret VaR under the parametric, historical simulation, and Monte Carlo simulation methods.
- 3.d.** Describe advantages and limitations of VaR.
- 3.e.** Describe extensions of VaR.
- 3.f.** describe sensitivity risk measures and scenario risk measures and compare these measures to VaR.
- 3.g.** demonstrate how equity, fixed-income, and options exposure measures may be used in measuring and managing market risk and volatility risk.
- 3.h** describe the use of sensitivity risk measures and scenario risk measures.
- 3.i.** describe advantages and limitations of sensitivity risk measures and scenario risk measures.

LEARNING OUTCOMES

- 3.j.** Explain constraints used in managing market risks, including risk budgeting, position limits, scenario limits, and stop-loss limits.
- 3.k.** Explain how risk measures may be used in capital allocation decisions.
- 3.l.** Describe risk measures used by banks, asset managers, pension funds, and insurers.

[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

1.

Value at Risk (VaR)

Definition

The **minimum loss** that would be expected a certain **percentage of the time** over a certain **period of time** given the assumed market conditions.

VaR has three components:

VaR

The loss size (minimum loss)

The probability (percentage of the time)

The time frame (period of time)

[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

1.

Value at Risk (VaR)

Consider the following statement:

*The **5%** VaR of a portfolio is **€2.2 million** over a **one-day period**.*

In the statement, the three component that were mentioned are:

€2.2 million

This is the **minimum loss** expected. VaR can be measured in either currency units (as in example) or in percentage terms.

For a **€400 million** portfolio, the VaR expressed in percentage term would be:

*The **5%** VaR of a portfolio is **0.55%** over a **one-day period**.*

5%

This is the **probability** of loss over **€2.2 million'** occurrence. 5% means that every 20 period, we expect there is 1 period that minimum loss is at least €2.2 million.

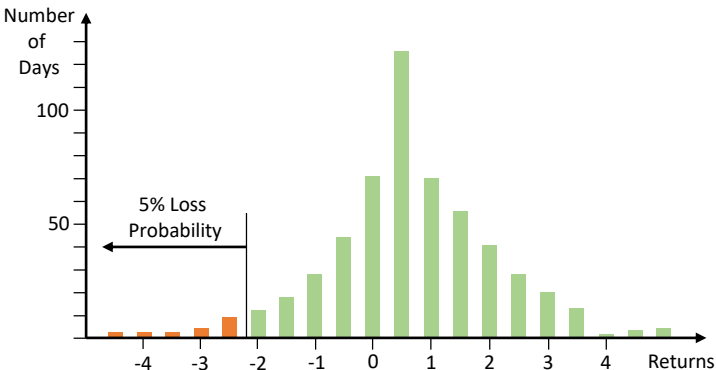
**one-day
period**

This is the **time period** of VaR. Each day has 5% chance that minimum loss is at least 5%

[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

1. Value at Risk (VaR)

The **5%** VaR of a portfolio is **€2.2 million** over a **one-day period**.



It is also correct to say any of the following:

- €2.2 million is the minimum loss we would expect 5% of the time.
- 5% of the time, losses would be at least €2.2 million.
- We would expect a loss of no more than €2.2 million 95% of the time.

[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

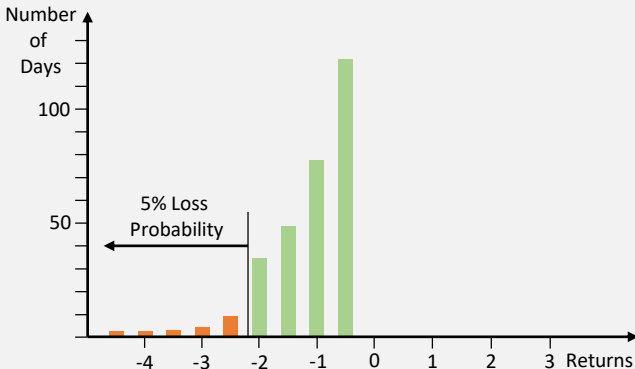
1. Value at Risk (VaR)

Note:

It is **wrong** to rephrase the example as:

*95% of the time we would expect to lose less than €2.2 million**

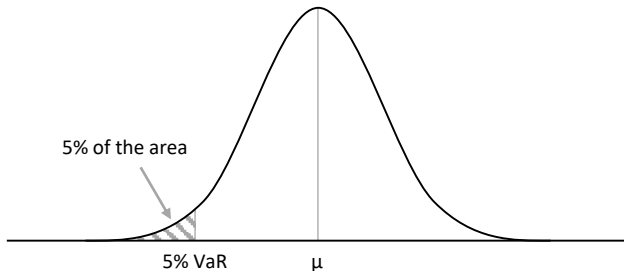
This statement implies that **we would always incur losses**, but only 5% of the time the loss is more than €2.2 million. The chart then would look like this:



[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

1. Value at Risk (VaR)

Illustration of 5% VaR in the Context of a Probability Distribution



Examination of the probability curve:

- The distribution is centered on μ (expected value). This is the value expected to happen most frequently.
- The 5% of the data points are within the shaded area under the curve, meaning that 5% of the time, portfolio's returns are expected to fall to the left of the 5% VaR point.

→ VaR is a point on the probability distribution of returns from a portfolio.

[LOS 3.a] Explain the use of value at risk (VaR) in measuring portfolio risk

1.

Value at Risk (VaR)



Knowledge check

Example: Definition of VaR

Given a VaR of \$12.5 million at 5% for one month, which of the following statements is correct?

- A. There is a 5% chance of losing \$12.5 million over one month.
- B. There is a 95% chance that the expected loss over the next month is less than \$12.5 million.
- C. The minimum loss that would be expected to occur over one month 5% of the time is \$12.5 million.

[LOS 3.a] *Explain* the use of value at risk (VaR) in measuring portfolio risk

1.

Value at Risk (VaR)



Knowledge check

Example: Definition of VaR

Given a VaR of \$12.5 million at 5% for one month, which of the following statements is correct?

- A. There is a 5% chance of losing \$12.5 million over one month.
- B. There is a 95% chance that the expected loss over the next month is less than \$12.5 million.
- C. The minimum loss that would be expected to occur over one month 5% of the time is \$12.5 million.

Answer

- A. There is a 5% chance of losing \$12.5 million over one month.
→ This is incorrect, because VaR does not give the likelihood of losing a specific amount.
- B. There is a 95% chance that the expected loss over the next month is less than \$12.5 million.
→ This is incorrect, because VaR is not an expected loss. It is a minimum loss.
- TC is correct, this is the only statement that accurately expresses the VaR.

[LOS 3.b, c] *Compare, estimate and interpret* VaR under the parametric, historical simulation, and Monte Carlo simulation methods

1.

Estimating VaR – Common Framework and Assumptions

Methods to estimate VaR

This module covers three methods that are typically used to estimate VaR:

- The parametric (variance-covariance) method
- The historical simulation method
- The Monte Carlo Simulation method

General steps in estimating VaR process

Risk
decomposition



Gathering data
history



Make an estimate
of the VaR

With the objective is to understand the concept of VaR, be aware of how it is estimated, know how it is used, we will work with a simplified portfolio. The portfolio is consist of two ETFs:

- ETFs—SPDR S&P 500 ETF (SPY), representing the US equity exposure
- SPDR Portfolio Long-Term Corporate Bond ETF (SPLB), representing a corporate bond exposure.

→ *The detailed portfolio is shown in the next slide*

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

1.

Estimating VaR – Common Framework and Assumptions

The portfolio we are using as input for this module is as follow:

		Annualized	
		Return	Standard Deviation
SPY (Asset A - stock)	80%	10.5%	20.0%
SPLB (Asset B – bond)	20%	6.0%	8.5%

- The correlation of A and B = -0.06
- Total value of the portfolio: \$150,000,000

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

Definition

The parametric (variance-covariance) method uses the estimated variances and covariances of portfolio securities to estimate the distribution of possible portfolio values, *often assuming a normal distribution*.

Note:

This method *typically* uses normal distribution. That is the normal case in practice, but it is *not required* to use normal distribution.

- The normal distribution is convenient because it only requires two parameters: the expected value and standard deviation.
- More complex distributions require additional parameters, such as skewness and kurtosis.

→ We limit the presentation in this module to normal distribution, because more accurate distributions would add complexity to the VaR estimation process.

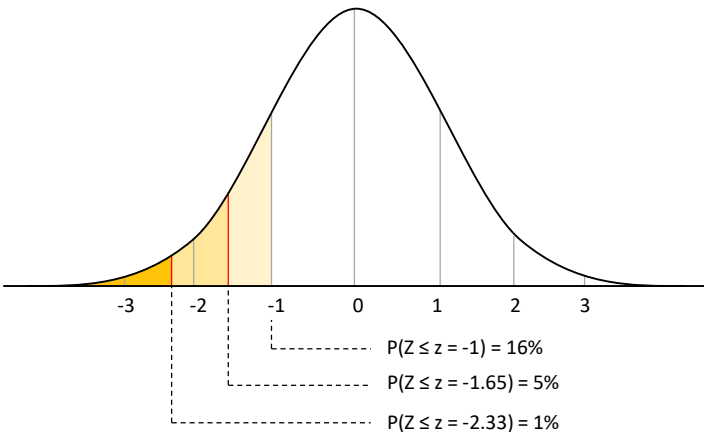
[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.1. z-distribution

A standardized distribution (z-distribution) is a standardized normal distribution with the expected value equals zero and the standard deviation equals one.



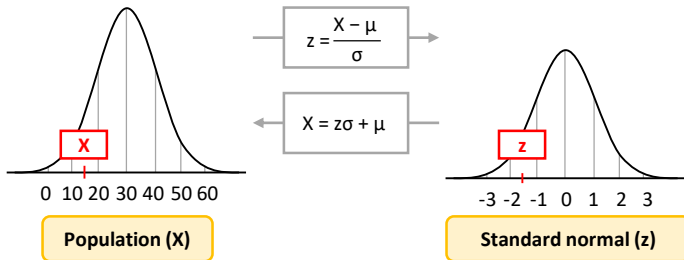
[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.2. Standardization of Normal Distribution

A **normal distribution** with **expected value μ** and **standard deviation σ** can be converted to a **standard normal distribution (z-distribution)** via the following formula:



→ The standard normal distribution shows that a value X with $P(x \leq X)$ is z standard deviations away from the expected value.

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.2. Standardization of Normal Distribution

For a given value z or a given probability of $P(X \leq x)$, we can identify the remaining of the two using the z -table.

Exhibit: A portion of z -table

z	0.00	0.01	0.02	0.03
2.0	0.9773	0.9778	0.9783	0.9788
2.1	0.9821	0.9826	0.9830	0.9834
2.2	0.9861	0.9865	0.9868	0.9871
2.3	0.9893	0.9896	0.9898	0.9901

We have worked with the z -table in Quantitative Method – CFA level 1. The z -table illustrates how many percent of the datapoints lies below a z -point.

Example: 99% of the datapoints lie below the 2.33 point on the z -distribution chart.

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

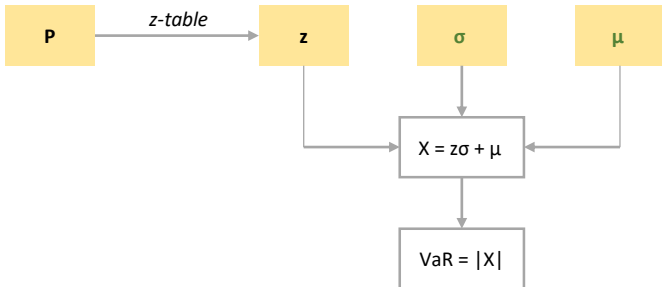
2.

The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

Recall that in defining VaR, we identified a VaR threshold – a point in the left tail of the distribution that $P(x \leq \text{VaR}) =$ the specified probability of VaR.

- The process of finding VaR value starts with using the prespecified probability by looking up in the z-table to find z-value.
- Now as we know the z-value, the VaR threshold on the normal distribution can be calculated using “ μ ” and “ σ ”.



[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

The previous parts shows that we need expected return and volatility of the portfolio to estimate VaR.

Expected return of the portfolio

$$E(R_p) = w_A E(R_A) + w_B E(R_B)$$

Volatility of the portfolio

$$\sigma_p = \sqrt{w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho_{A,B} \sigma_A \sigma_B}$$

Where:

- $E(R)$ = Expected return
- w = weight
- σ = volatility (standard deviation)
- ρ = correlation of returns of A and B

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

The portfolio we are using as input for this module is as follow:

		Annualized	
	Allocation	Return	Standard Deviation
SPY (Asset A - stock)	80%	10.5%	20.0%
SPLB (Asset B – bond)	20%	6.0%	8.5%

- The correlation of A and B = -0.06
- Total value of the portfolio: \$150,000,000

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

Base on assumptions of inputs, these estimates are calculated as follows:

$$E(R_p) = 0.8(0.105) + 0.2(0.06) = 0.096$$

$$\begin{aligned}\sigma_p &= \sqrt{(0.8)^2(0.2)^2 + (0.2)^2(0.085)^2 + 2(0.8)(0.2)(-0.06)(0.2)(0.085)} \\ &= 0.159883\end{aligned}$$

→ Thus, our portfolio, consisting of an 80% position in stocks and 20% position in bonds have an expected return of 9.6% and a volatility of approximately 15.99%

From annual calculation, we can calculate one-day VaR with a very important assumption: ***the statistical properties of the return distribution are constant across the year.***

$$E(R_p) = \frac{0.096}{250} = 0.000384$$

$$\sigma_p = \frac{0.159883}{\sqrt{250}} = 0.010112$$

[LOS 3.b, c] Compare, estimate and interpret VaR ...**2.****The Parametric Method of VaR Estimation****2.3. Estimating VaR Using Parametric Method****Note:**

The conversion from annual VaR to daily VaR is an **one way path**. We cannot estimate a daily VaR and annualize it to arrive at an annual VaR estimate. There are two reasons for this:

- First, to assume that a daily distribution of returns can be extrapolated to an annual distribution is a bold assumption.
- Second, annualizing the daily VaR is not the same as equalizing the parameters. We can annualize the daily return and the standard deviation and estimate an annual VaR with additional assumptions.

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2. The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

Now, we have collected sufficient inputs to estimate daily VaR using Parametric Method:

- Daily expected return (μ): $E(R_p) = 0.000384$
- Daily standard deviation: $\sigma_p = 0.010112$
- The probability of VaR

The following step-by-step procedure shows how the 5% daily VaR is derived:

1 Find the z value using the z table

The z-value of 5% is -1.65

2 Calculate X using z, expected return and standard deviation

$$X = z\sigma + \mu = -1.65 \times 0.010112 + 0.000384 = -0.016301$$

3 Express VaR as an absolute number (despite representing an expected loss)

$$5\% \text{ VaR} = |-0.016301| = 0.016301$$

4 Multiply the result in Step 3 by the value of the portfolio

$$\$150,000,000 \times 0.016301 = \$2,445,150$$

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation

2.3. Estimating VaR Using Parametric Method

Thus, using the parametric method, our estimate of VaR is \$2,445,150, meaning that on 5% of trading days the portfolio would be expected to incur a loss of at least \$2,445,150.

Or we can say:

- *\$2,445,150 is the minimum loss we would expect 5% of the time.*
- *5% of the time, losses would be at least \$2,445,150.*
- *We would expect a loss of no more than \$2,445,150 95% of the time.*

2.3. Advantages and Disadvantages of Parametric VaR

Advantages

Simple to apply under the assumption of normally distributed returns.

Disadvantages

Limited usefulness when the portfolio contains options

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation



Knowledge check

Example: Parametric VaR

Assuming a daily expected return of 0.0384% and daily standard deviation of 1.0112% (as in the example in the text), which of the following is closest to the 1% VaR for a \$150 million portfolio? Express your answer in dollars.

- A. \$3.5 million
- B. \$2.4 million
- C. \$1.4 million

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

2.

The Parametric Method of VaR Estimation



Knowledge check

Example: Parametric VaR

Assuming a daily expected return of 0.0384% and daily standard deviation of 1.0112% (as in the example in the text), which of the following is closest to the 1% VaR for a \$150 million portfolio? Express your answer in dollars.

- A. \$3.5 million
- B. \$2.4 million
- C. \$1.4 million

Answer:

A is correct and is obtained as follows:

Step 1: The z-value of 1% VaR is 2.33

Step 2: Calculate X: $X = -2.33 \times 0.010112 + 0.000384 = -0.023177$

Step 3: $|-0.023177| = 0.023177$

Step 4: $0.023177 \times \$150 \text{ million} = \$3,476,550$

B is the estimated VaR at a 5% threshold, and C is the estimated VaR using a one standard deviation threshold.

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

3.

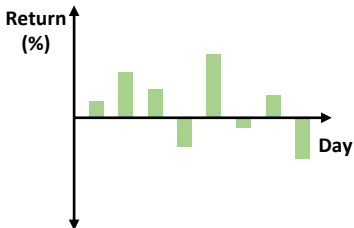
The Historical Simulation Method of VaR Estimation

Definition

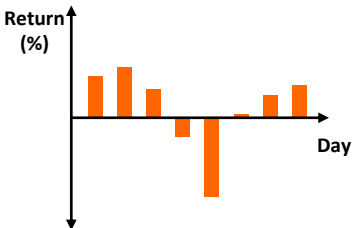
Historical simulation method of VaR uses the current portfolio and reprices it using the actual historical changes in the key factors experienced during the lookback period.

Estimating VaR Using Parametric Method

- 1 *Gathering the historical returns of each risk factor from the chosen look back period*



Return of A



Return of B

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

3.

The Historical Simulation Method of VaR Estimation

Estimating VaR Using Parametric Method

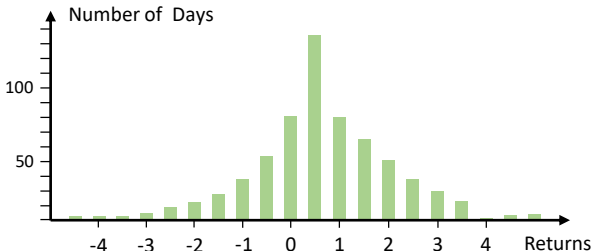
2 *Reprice the current portfolio using the returns data from the look back period*

The portfolio we are working with consist of 80% asset A and 20% of asset B. To come up with the portfolio returns, we multiply the return of each holding by its respective weight in the port folio and add the result together.

Example: On 1/1/20XY, the return of asset A is 0.8% and asset B is -0.53%.

Return of the portfolio on 1/1/20XY = $0.8(0.008) + 0.2(-0.0053) = 0.53\%$

3 *Sort the result from largest loss to greatest gain*



[LOS 3.b, c] Compare, estimate and interpret VaR ...

3.

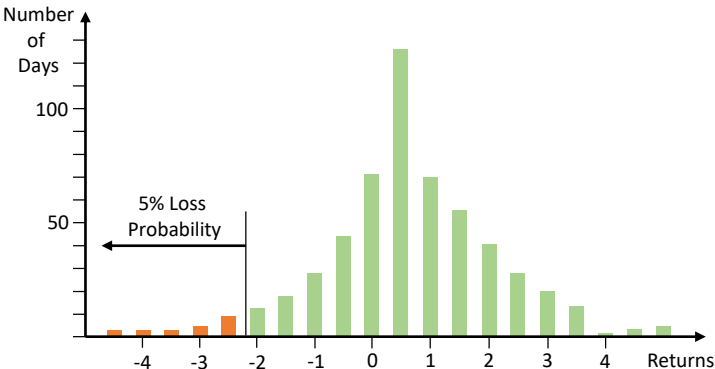
The Historical Simulation Method of VaR Estimation

Estimating VaR Using Parametric Method

4

Choose the point on the resulting distribution that matches our desiring VaR

To estimate a one-day VaR at a 5% confidence interval, we choose the point on the resulting distribution beyond which 5% of the outcomes result in larger losses.



[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

3.

The Monte Carlo Simulation Method of VaR Estimation

Definition

Monte Carlo simulation develops assumptions about the statistical characteristics of the distribution and uses those characteristics to generate random outcomes that represent hypothetical returns to a portfolio with the specified characteristics.

Monte Carlo Simulation work is widely used in complex situations

**The estimation
has many
uncertain
variables**

**Complex
portfolios consist
of options**

**Portfolios with
large amount of
assets and risk
factors**

Note: The application of Monte Carlo Simulation is quite similar to Historical Simulation Method. The main difference is in Monte Carlo Simulation, we use computer to generate a much larger amount of sample outcomes.

[LOS 3.b, c] *Compare, estimate and interpret VaR ...*

3.

The Monte Carlo Simulation Method of VaR Estimation

Estimating VaR Using Monte Carlo Simulation

1 *Establish the mathematical model*

Define an equation that brings the output and input variables together.

2 *Determine the input values*

In our example, our input values are returns of SPY (asset A) and SPLB (asset B).

3 *Create a sample dataset*

We generate the 10,000 outcomes of which all the returns are paired and weighted 80/20.

4 *Sort the outcomes from worst to best and select the point for desiring VaR*

This step is similar to historical simulation method

[LOS 3.d] *Describe* advantages and limitations of VaR

1. Advantages of VaR

1. *Simple concept*

VaR is easy to understand. While the methodology is fairly technical, the concept itself is not very difficult.

→ Decision makers without technical backgrounds should be able to grasp the likelihood of possible losses that might endanger the organization.

2. *Easily communicated concept*

VaR captures a considerable amount of information into a single number.

→ VaR itself can be a very significant and practical piece of information.

3. *Provides a basis for risk comparison*

VaR can be useful in comparing risks across asset classes, portfolios, and trading units.

This point will be discussed further in a later section.

[LOS 3.d] *Describe* advantages and limitations of VaR

1. Advantages of VaR

4. *Facilitates capital allocation decisions*

The ability to compare VaR across trading units or portfolio positions provides management with a benchmark that can be used in capital allocation decisions.

This point will be discussed further in a later section.

5. *Can be used for performance evaluation*

Risk-adjusted performance measurement requires that return or profit be adjusted by the level of risk taken. VaR can serve as the basis for risk adjustment.

6. *Reliability can be verified*

VaR is easily capable of being verified, a process known as backtesting – using historical data to determine VaR's reliability.

7. *Widely accepted by regulators*

Many regulators require or encourage the use of VaR in disclosing risk of financial assets.

[LOS 3.d] *Describe* advantages and limitations of VaR

2. Limitations of VaR

1. *Subjectivity*

VaR is actually a rather subjective method, for its dependency on various input assumptions.

2. *Underestimating the frequency of extreme events*

In other words, there are often more extreme adverse events, called “left-tail events,” than would be expected under a normal distribution.

3. *Failure to take into account liquidity*

If some assets in a portfolio are relatively illiquid, VaR could be understated, even under normal market conditions.

4. *Sensitivity to correlation risk*

Correlation risk is the risk that during times of extreme market stress, correlations among all assets tend to rise significantly.

[LOS 3.d] *Describe* advantages and limitations of VaR

2.

Limitations of VaR

5.

Misunderstanding the meaning of VaR

VaR is not a worst-case scenario. Losses can and will exceed VaR.

6.

Oversimplification

Although we noted that VaR is an easily communicated concept, it can also oversimplify the picture.

7.

Disregard of right-tail events

VaR focuses so heavily on the left tail (the losses) that the right tail (potential gains) are often ignored.

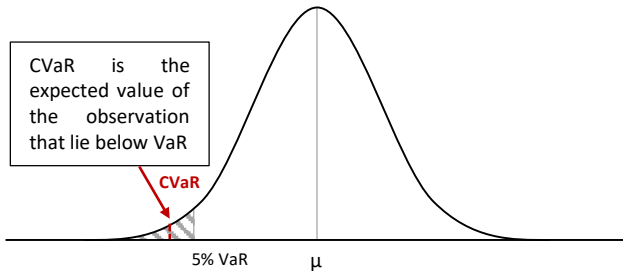
[LOS 3.e] *Describe extensions of VaR*

1.

Conditional VaR (CVaR)

Definition

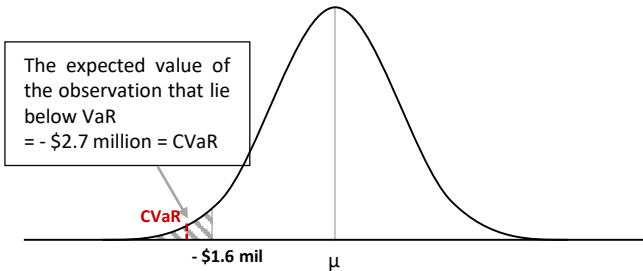
The expected loss, given that the loss is equal to or greater than the VaR.



[LOS 3.e] *Describe extensions of VaR*

1.

Conditional VaR (CVaR)



Example: Conditional VaR

In a historical simulation estimation of VaR, the sample of 1,000 historical returns was sorted from lowest to highest and the 5% VaR was \$1.6 million. There are 50 observations (5% of 1,000) lie below the VaR estimate, and the average of these loss is \$2.7 million.

→ When VaR is exceeded, we would expect an average loss of about \$2.7 million.

[LOS 3.e] *Describe extensions of VaR*

2.

Incremental VaR (IVaR)

Definition

The change in VaR of the portfolio when a position size is changed relative to the remaining positions.

A change in position size results in a change in estimated VaR

	Weight
Asset A	80%
Asset B	20%
→ Estimated VaR(1)	

	Weight
Asset A	90%
Asset B	10%
→ Estimated VaR(2)	

IVaR

$$\text{IVaR} = \text{Estimated VaR}(2) - \text{Estimated VaR}(1)$$

Example: Incremental VaR (IVaR)

If a 2% increase in the weight of a security in the portfolio increases the portfolio's VaR from \$1,345,600 to \$1,562,400.

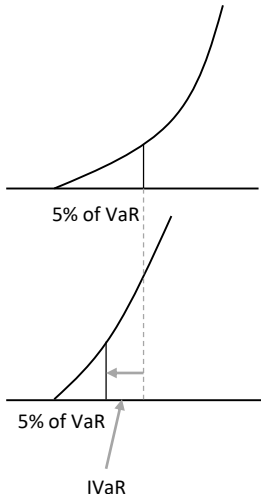
→ The IVaR for the 2% increase in the portfolio weight of the security is $1,562,400 - 1,345,600 = \$216,800$.

[LOS 3.e] *Describe extensions of VaR*

2.

Incremental VaR (IVaR)

Illustration: A change in position size results in a change in estimated VaR



The original return distribution and VaR

Position size changes

The distribution are reshaped

The left tail is fatter and more area is distributed on the left tail.

→ The 5% VaR shifted

[LOS 3.e] *Describe extensions of VaR*

3.

Marginal VaR (MVar)

Definition

The change in VaR of the portfolio that reflects the effect of a very small change in the position.

→ This is a similar concept to IVaR, only the change in position of MVar is very small (marginal)

4.

Ex ante tracking error (relative VaR)

Definition

The VaR of the difference between the return on a portfolio and the return on its manager's benchmark portfolio.

Example: Ex ante tracking error (relative VaR)

A 5% monthly relative VaR of 2.5% implies that 5% of time, the portfolio's relative underperformance will be at least 2.5%. The relative VaR can be calculated as the VaR of a combination of a long position in the subject portfolio and a short position in the benchmark portfolio.

[LOS 3.e] *Describe* sensitivity risk measures and scenario risk measures and compare these measures to VaR

1.

Sensitivity Risk Measures

Definition

Sensitivity measures examine how performance responds to a single change in an underlying risk factor.

1.1. Equity Exposure Measures - Beta

Capital Asset Pricing Model (CAPM)

$$E(R_i) = R_F + \beta_i[E(R_M) - R_F]$$

- In this often-used equation, β measures the sensitivity of the security's expected return to the equity risk premium.
- The broad market beta, which is an average of all individual betas, is 1.0. Assets with betas more (less) than 1 are considered more (less) volatile than the market as a whole.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

1.

Sensitivity Risk Measures

1.2. Fixed-Income Exposure Measures – Duration & Convexity

The **duration** is a measure of the sensitivity of the bond price to the interest rate change that characterizes all rates.

Duration

$$\frac{\Delta B}{B} \approx -D \frac{\Delta y}{1+y'}$$

Where:

- D: Duration of the bond
- Y: yield of the bond
- B: Bond price

To accommodate longer periods of time and larger yield change, we incorporate **convexity**

Duration & Convexity

$$\frac{\Delta B}{B} \approx -D \frac{\Delta y}{1+y'} + \frac{1}{2} C \frac{y^2}{(1+y)^2}$$

Where:

- C: Convexity of the bond

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

1.

Sensitivity Risk Measures

1.3. Option Risk Measures

Delta

Delta is a measure of the sensitivity of the option value to underlying asset price change.

Delta

$$\Delta \approx \frac{\text{Change in value of option}}{\text{Change in value of underlying}}$$

Recall that call options have limited downside, while put option have limited upside (to the buyer):

- Call option deltas range from 0 to 1;
- Put option deltas range from -1 to 0;

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

1.

Sensitivity Risk Measures

1.3. Option Risk Measures

Delta

Types of option	Value of Delta	Meaning
Call	0	The option value does not change when the value of the underlying changes.
	1	Price of the call option changes in unision with the underlying
Put	0	The option value does not change when the value of the underlying changes (similar to the call).
	-1	Price of the call option changes in unision with the underlying, but in opposite direction.

Note:

The extreme value of delta (-1; 0; 1) are conditions that is never absolutely true, but:

- 0 can be roughly true for very deep out-of-the-money options.
- -1; 1 can be roughly true for very deep in-the-money put/call options.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

1.

Sensitivity Risk Measures

1.3. Option Risk Measures

Gamma

- Similar to duration, which has the second-order effect of convexity, we can add a second-order effect for options called **gamma**.
- Gamma measures how Delta changes when the value of the underlying changes.

Gamma

$$\Gamma \approx \frac{\text{Change in delta}}{\text{Change in value of underlying}}$$

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

1.

Sensitivity Risk Measures

1.3. Option Risk Measures

Vega

- A third important sensitivity measure for options is **vega**, and it reflects the effect of volatility to the value of the option.

Vega

$$\text{Vega} \approx \frac{\text{Change in value of option}}{\text{Change in volatility of underlying}}$$

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures

Definition

Scenario Analysis is the method of estimating the effect on portfolio value made by a set of changes of significant magnitude in multiple risk factor.

2.1. Historical Scenario

A **historical scenario** approach uses a set of changes in risk factors that have actually occurred in the past, especially changes during a period of financial disruption and stress.

**Current
portfolio**

***A condition that
happened in the past***

**A scenario
outcome of
current
portfolio**

The scenarios chosen in historical scenario approach are usually crisis periods such as subprime mortgage crisis of 2008 or the equity market crash of 1987. *We will discuss the purpose further in the coming section (2.3)*

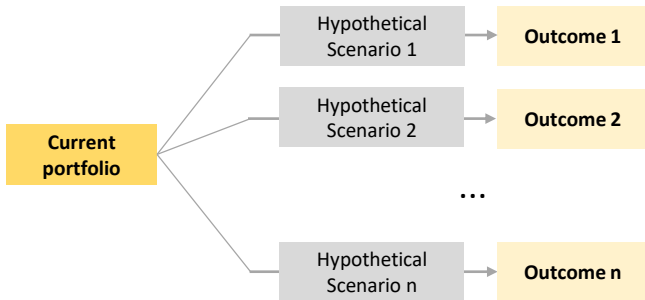
[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures

2.2. Hypothetical Scenario

A **hypothetical scenario** approach uses any set of changes in risk factors (not necessarily occurred in the past)



Note:

Because of the limitless assumptions in factors' changes, a hypothetical scenario could have more extreme changes in risk factors than those that have occurred in the past.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures

2.3. Stress tests

Definition

The process of applying extreme negative stress to a particular portfolio exposure.

Key difference between stress tests and scenario analysis:

- Scenario analysis looks at all the possible outcome, positive or negative;
- Stress tests focus on extreme negative events to assess the impact of such an event on the portfolio (focus on left-tail risks).

→ Stress tests support scenario analysis by a process called **reverse stress testing**.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

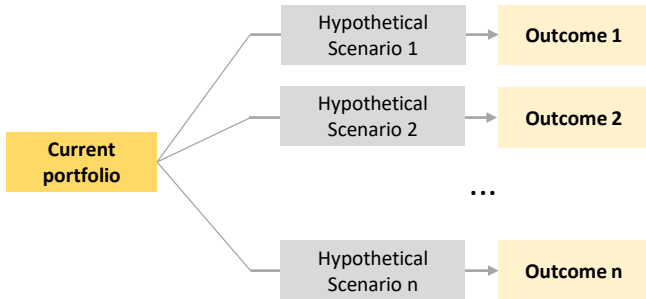
Scenario Risk Measures

2.4. Reverse stress testing

Definition

The process of targeting portfolio's material exposures and assessing their behavior in various environment.

To understand the reverse stress testing framework, we first take a look at the hypothetical approach in scenario analysis



As the name of the method, we basically reverse the steps:

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures

2.4. Reverse stress testing

- 1** *Identify a portfolio's largest risk exposures.*
- 2** *Determine unacceptable outcomes*
- 3** *Identify scenarios that changes in risk factors that would result in such outcomes*
- 4** *Estimating the probability of undesired scenario happening*

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

1. Which of the following is an example of a reverse stress test?
 - A. Identify the top 10 exposures in the portfolio, and then generate a hypothetical stress that could adversely affect all 10 simultaneously.
 - B. Find the worst single day's performance that could have occurred for the current portfolio had it been held throughout the past five years.
 - C. Find the returns that occurred in all risk factors in the 2008 global financial crisis, reverse the sign on these, and apply them to today's portfolio.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

1. Which of the following is an example of a reverse stress test?
 - A. Identify the top 10 exposures in the portfolio, and then generate a hypothetical stress that could adversely affect all 10 simultaneously.
 - B. Find the worst single day's performance that could have occurred for the current portfolio had it been held throughout the past five years.
 - C. Find the returns that occurred in all risk factors in the 2008 global financial crisis, reverse the sign on these, and apply them to today's portfolio.

Answer:

- A is correct.
- B is not a reverse stress test because reverse stress tests focus more narrowly on trouble spots for a specific portfolio.
- C would illustrate how the portfolio would have performed in an extremely strong market, quite unlike what occurred in 2008.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

2. Which kind of market participant is least likely to use scenario analysis as a pass/fail stress test?

- A. Bank
- B. Long-only asset manager
- C. Hedge fund using leverage

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

2. Which kind of market participant is least likely to use scenario analysis as a pass/fail stress test?

- A. Bank
- B. Long-only asset manager
- C. Hedge fund using leverage

Answer:

- B is correct. Long-only asset managers do not typically use leverage and are thus less likely to become insolvent, making a pass/fail test for solvency less relevant to them.
- A and C are not correct because parties that use leverage, such as hedge funds and banks, are likely to use stress tests to determine what market movements could impair their capital and lead to insolvency.

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

3. Which of the following is not a limitation of scenario measures?

- A. It is difficult to ascribe probability to a given scenario
- B. Scenario measures assume a normal distribution, and market returns are not necessarily normal
- C. They risk being an infinite task; one cannot possibly measure all of the possible future scenarios

[LOS 3.f-h] *Describe* sensitivity risk measures and scenario risk measures, compare these measures to VaR and uses of these measures

2.

Scenario Risk Measures



Knowledge check

Example: Scenario Analysis

3. Which of the following is not a limitation of scenario measures?

- A. It is difficult to ascribe probability to a given scenario
- B. Scenario measures assume a normal distribution, and market returns are not necessarily normal
- C. They risk being an infinite task; one cannot possibly measure all of the possible future scenarios

Answer:

- B is correct. Scenario measures do not assume any given distribution, and thus this is not a limitation of scenario analysis.
- A is incorrect because it is in fact difficult to ascribe probability to many scenarios, and thus this is a limitation of scenario analysis.
- C is also incorrect because it is in fact impossible to measure all possible future scenarios, and this is a limitation of scenario analysis.

[LOS 3.g] Describe advantages and limitations of sensitivity risk measures and scenario risk measures

Var, sensitivity analysis, and scenario analysis complement each other, and a risk manager should not rely on only one of these measures.

Ways that sensitivity and scenario risk measures can complement Var:

- They do **not always need to rely on history**. Sensitivity and scenario risk measures can be constructed to test the portfolio's vulnerability to a truly never-before-seen market movement.
- Scenarios can be designed to **overcome any assumption of normal distributions**.
- Scenarios can **be tailored to expose a portfolio's most concentrated positions** to even worse movement than its other exposures, allowing liquidity to be taken into account.

However, they have some limitations as well:

- Historical scenarios are interesting, and illuminating, but are **not going to happen in exactly the same** way again.
- Hypothetical scenarios may **incorrectly** specify how assets will co-move, get the magnitude of movements wrong, or incorrectly adjust for the effects of liquidity and concentration.
- Hypothetical scenarios can be very **difficult to create and maintain**.
- It is very difficult to know **how to establish the appropriate limits** on a scenario analysis or stress test

[LOS 3.j] Explain constraints used in managing market risks, including risk budgeting, position limits, scenario limits, and stop-loss limits.

Risk limits that are often imposed include the following.

**Risk
budgeting**

The process: Determine the acceptable total risk for an organization → Allocate that risk to different activities, strategies, or asset classes as appropriate.

Example: *Determining the maximum allowable 5% VaR amount, then allocating that VaR across various business units.*

**Position
limits**

Ensure some minimum level of diversification by limiting risk exposures.

Example: *position limits on allocations to individual securities within an asset class, on asset classes such as equities or high-yield bonds,...*

**Scenario
limits**

Are limits on expected loss for a given scenario.

**Stop-loss
limits**

Require that a risk exposure be reduced if losses exceed a specified amount over a certain period of time.

Example: *Requirement to reduce the portfolio allocation to a stock (by a given amount) if it declines in value by more than a specified percentage.*

[LOS 3.k] Explain how risk measures may be used in capital allocation decisions.

Capital Allocation

The practice of placing limits on each of a company's activities in order to ensure that: the areas in which it expects the greatest reward and has the greatest expertise are given the resources needed to accomplish their goals.

Capital Allocation Process

Measure economic capital

The amount of capital a firm needs to hold if it is to survive severe losses from the risks in its businesses.

**Establishes overall risk appetite in
economic capital terms**

**Subdivide this appetite among its
units**

[LOS 3.I] *Describe* risk measures used by banks, asset managers, pension funds, and insurers.

Financial institutions

Risk measures

Banks

- Liquidity gap (*liquidity and asset/liability mismatch*)
- VaR (*for the held-for-sale or trading (fair value) securities*)
- Leverage (*risk assets divided by equity*)
- Sensitivities (*for the held-for-sale securities*)
- Economic capital (*the total loss the company could suffer*)
- Scenario analysis (*stress tests and others*)

Asset Managers

- Backward-looking returns-based measures (*standard deviation, ex post tracking error, Sharpe ratio, information ratio, and historical beta*)
- Risk exposure measures (*VaR, ex ante tracking error, duration and forward-looking beta, stress tests, and scenario analyses*)

Traditional Asset Managers

- Position limits
- Sensitivities (*option-adjusted duration, key rate duration, and credit spread duration, delta exposure of options*)
- Beta sensitivity (*for equity-only accounts*)
- Liquidity
- Scenario analysis
- Redemption risk (*percentage of the portfolio could be redeemed at peak times*)
- Ex post versus ex ante tracking error
- VaR

[LOS 3.I] *Describe* risk measures used by banks, asset managers, pension funds, and insurers.

Financial institutions

Risk measures

Hedge Funds

- Sensitivities
- Gross exposure (the sum of the absolute value of long plus short positions)
- Leverage
- VaR
- Scenarios/ stress tests
- Drawdown (*maximum drawdown*)

Pension funds

- Interest rate and curve risk
- Surplus at risk (how much the assets might underperform the liabilities)
- Liability hedging exposures versus return generating exposures

Insurers

Property and casualty insurance:

- Sensitivities and exposures
- Economic capital and VaR
- Scenario analysis

Life insurance:

- Sensitivities
- Asset and liability matching
- Scenario analysis



MODULE 4: BACKTESTING AND SIMULATION

LEARNING OUTCOMES

- 4.a.** Describe objectives in backtesting an investment strategy
- 4.b.** Describe and contrast steps and procedures in backtesting an investment strategy
- 4.c.** Interpret metrics and visuals reported in a backtest of an investment strategy
- 4.d.** Identify problems in a backtest of an investment strategy
- 4.e.** Evaluate and interpret a historical scenario analysis
- 4.f.** Contrast Monte Carlo and historical simulation approaches
- 4.g.** Explain inputs and decisions in simulation and interpret a simulation
- 4.h.** Demonstrate the use of sensitivity analysis

Overview

Backtesting

Backtesting objectives

Backtesting process

**Common problems in
backtesting**

**Historical scenario analysis
(Historical stress testing)**

Supplement

Analysis

Simulation analysis

Sensitivity analysis

**Monte
Carlo
simulation**

**Historical
simulation**

[LOS 4.a] Describe objectives in backtesting an investment strategy

1.

Backtesting

Backtesting is a process by which we use actual historical data to emulate the investment process and determine whether a particular investment strategy or technique would have delivered the expected excess returns.

Assumption of backtesting: the future will at least somewhat resemble history.

Backtesting process is intuitive, because it mimics how investing is accomplished in reality.

Formulating
investment
strategies



Testing those
ideas using
historical
information



Assessing how
well those
strategies
would have
performed

[LOS 4.a] Describe objectives in backtesting an investment strategy

2. Objectives in backtesting

The primary goal of backtesting is *to assess the risk and return of an investment strategy* by simulating the investment process.

To reassure investors that investing strategies and models are likely to perform well.

To refine and improve the investors' investment process

While backtesting is a natural fit for systematic and quantitative investment styles, fundamental managers also make extensive use of backtesting.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

1.

Backtesting process

Backtesting consists of **three** steps:

- Strategy design
- Historical investment simulation
- Analysis of backtesting output

The exhibit below illustrates these steps and component procedures:

Backtesting Flowchart

Strategy
design

- Specify investment hypothesis and goal(s)
- Determine investment rules and process
- Decide key parameters

Historical
investment
simulation

- Form investment portfolios for each period according to the rules specified in the previous step
- Rebalance the portfolio periodically based on pre-determined rules

Analysis of
backtesting
output

- Calculate portfolio performance statistics
- Compute other key metrics (e.g., turnover, etc.)

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

2.

Step 1: Strategy Design

Strategy
design

- Specify investment hypothesis and goal(s)
- Determine investment rules and process
- Decide key parameters

The *first step* is to identify the investment **goals** and **hypothesis**.

- For active strategies, the **goal** is typically to achieve superior risk-adjusted absolute return.
- An investment **hypothesis** is a method (a trading rule, security selection criterion, a portfolio,...) aimed at achieving the goal.

The *next step* is to translate the hypothesis into rules and processes and to specify several key parameters.

Investment
universe

2.1

Specific
definition
of returns

2.2

Frequency
of portfolio
rebalancing

2.3

Start and
end dates

2.4

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

2.

Step 1: Strategy Design

2.1. Investment universe

The term **investment universe** is a reference to all the assets in which we might conceivably take a position while executing our strategy.

For example, we may use the VN-Index, or VN30 Index as our investment universe.

2.2. Specific definition of returns

Currency of return: If our investment universe extends beyond our own country's borders, a number of complications emerge such as *whether to denominate return in the local currency of the investment, or to translate returns into a single currency.*

Benchmark of return:

- Returns also should be reported relative to a particular benchmark.
- Typically, the benchmark will be an index that reflects the investment universe of our strategy.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

2. Step 1: Strategy Design

2.3. Frequency of portfolio rebalancing

Monthly rebalancing is typical for investment portfolios, though shorter or longer frequencies are not uncommon.

When higher-frequency rebalancing is used, *transaction costs* will increase and can become significant.

→ Performance presentation of a backtest should specify whether or not transaction costs are included.

2.4. Start and end dates

Portfolio managers generally prefer to use **a long data history** when backtesting strategies, in order to maximize confidence in the results.

BUT

Financial market data tends to be *nonstationary*, in that it is likely to contain various distinct regime.

The *discrete time intervals* within the history should additionally be *individually* analyzed.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

3.

Step 2: Historical Investment Simulation

Historical
investment
simulation

- Form investment portfolios for each period according to the rules specified in the previous step
- Rebalance the portfolio periodically based on pre-determined rules

The *first step* is to form investment portfolios for each period according to the rules specified .

- It depends primarily on the *investment hypothesis* under consideration, the investment *manager's capabilities and style*, and the *client's investment mandate* for which the potential strategy is relevant.
- Backtesting can be applied to any kind of investment strategy.

The *next step* is to rebalance the portfolio periodically based on pre-determined rules.

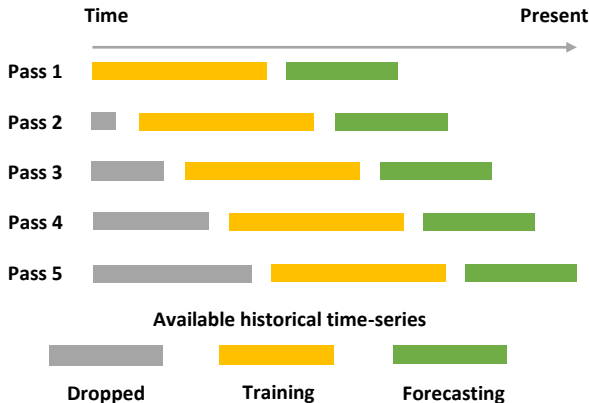
→ **Rolling window** is typically used to simulate rebalancing.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

3.

Step 2: Historical Investment Simulation

In the **rolling window backtesting** approach, a portfolio or strategy is constituted at the beginning of a period using data from a historical in-sample period, followed by testing on a subsequent, out-of-sample period.



The process is repeated as time moves forward. This approach replicates the live investing process, because investment managers adjust their positions as new information arrives.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

3. Step 2: Historical Investment Simulation

Example: Assume we backtest a value strategy by measuring its performance each month from December 2011 to May 2012. The process begins on 30 November 2011 by compiling every stock's trailing 12-month earnings yield.

The process of recording the investment results is repeated at the end of each subsequent month, by rebalancing the portfolio with refreshed trailing 12-month earnings yield data and measuring the results over the ensuing (OOS) month.

Rolling Window Backtesting of the Earnings Yield Factor

	2010-12	2011-01	2011-02	2011-03	2011-04	2011-05	2011-06	2011-07	2011-08	2011-09	2011-10	2011-11	2011-12	2012-01	2012-02	2012-03	2012-04	2012-05
11/30/2011									In-Sample (Last 12M EPS/Price)				OOS					
12/31/2011										In-Sample (Last 12M EPS/Price)			OOS					
1/31/2012											In-Sample (Last 12M EPS/Price)		OOS					
2/29/2012												In-Sample (Last 12M EPS/Price)		OOS				
3/31/2012													In-Sample (Last 12M EPS/Price)		OOS			
4/30/2012														In-Sample (Last 12M EPS/Price)		OOS		

Source: Wolfe Research Luo's QES.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

4.

Step 3: Analysis of Backtesting Output

Analysis of
backtesting
output

- Calculate portfolio performance statistics
- Compute other key metrics (e.g., turnover, etc.)

The *first step* is to calculate portfolio performance statistics for presentation and interpretation such as *average return* and *risk profile (volatility and downside risk)*

→ Analysts often use metrics such as the Sharpe ratio, the Sortino ratio, volatility, and maximum drawdown (*discussed later*).

The *next step* is to compute other key metrics (visual outputs) such as *time series of returns*, *normal distribution of returns*, *cumulative performance* of an investment strategy over an extended history.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5.

Backtesting multifactor models

In this section, we introduce *two* multifactor equity portfolio strategies to more richly illustrate backtesting:

- A benchmark (BM) factor portfolio, which equally weights multiple fundamental factors.
- A risk parity (RP) factor portfolio, which weights factors based on equal risk contribution.

We chose these two approaches because their weighting schemes—equal weights and equal risk weights, respectively—are objective.

*To backtest these two portfolios, we follow the same three steps described previously: **strategy design, historical investment simulation, and output analysis.***

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5. Backtesting multifactor models

5.1. Step 1: Strategy design

We chose **eight** fundamental factors from common investment styles:

Investment style	Common factor	Which is preferred?
Defensive value	Trailing earnings yield	High E/P ratio
Cyclical value	Book-to-market ratio	High B/P ratio
Growth	Consensus earnings growth for the coming year	High earnings growth
Price momentum	12-month total return, excluding the previous month	Momentum
Analyst sentiment	EPS changes over the past 3 months	Positive earnings revisions

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5. Backtesting multifactor models

5.1. Step 1: Strategy design

We chose **eight** fundamental factors from common investment styles:

Investment style	Common factor	Which is preferred?
Profitability	Return on equity	High ROE
Leverage	Debt/equity ratio	Low leverage
Earnings quality	Level of accruals	Low levels of accruals

- **With the benchmark (BM) factor portfolio:** For each factor, we form a portfolio by buying the top 20% of stocks and shorting the bottom 20% of stocks ranked by the factor. Stocks within both long and short buckets are equally weighted.
- **With the risk parity (RP) factor portfolio:** We combine the eight factor portfolios by equally weighting them by their risk contribution. → It requires a complete variance – covariance matrix at each rebalancing date.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5.

Backtesting multifactor models

5.2. Step 2: Historical investment simulation

Backtesting a multifactor strategy is similar to the method introduced earlier, but the **rolling-window procedure** is implemented **twice**: once at the **factor level**, once at factor **portfolio level**.

Factor level: We construct the eight factor portfolios using the walk-forward approach.

Portfolio level: We use either the benchmark (**BM**) or risk parity (**RP**) weighting schemes to form a multifactor portfolio.

BM factor portfolio

Rebalance back to equal weight

RP factor portfolio

At each month end, a variance–covariance matrix must be re-estimated to rebalance back to risk parity

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

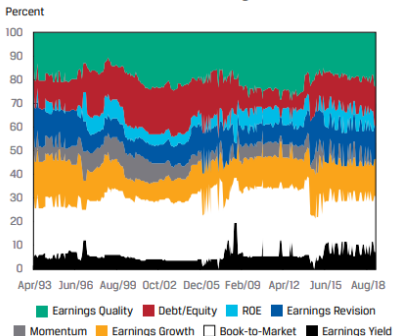
5.

Backtesting multifactor models

5.3. Step 3: Output Analysis

Let's see the exhibit below. It shows that the weights of the eight factor portfolios in the RP portfolio are relatively stable over time (1993–2019) in the United States.

A. RP Portfolio Allocation Weights in the US

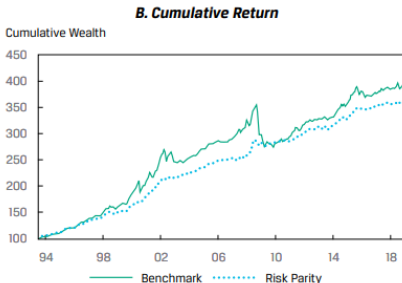


Notably, book-to-market and earnings quality factor portfolios receive the largest allocations, whereas ROE and price momentum factor portfolios have the lowest weights.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5. Backtesting multifactor models

5.3. Step 3: Output Analysis



Because the weights of the eight factor portfolios are certainly not equal—so we should expect the RP portfolio's risk and return profile to differ from that of the BM portfolio.

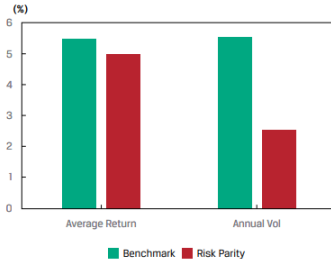
→ Clearly, the exhibit above shows that the RP portfolio appears to deliver a lower cumulative return than does the BM portfolio.

[LOS 4.b] Describe and contrast steps and procedures in backtesting an investment strategy

5. Backtesting multifactor models

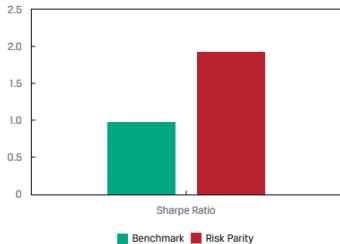
5.3. Step 3: Output Analysis

C. Average Return and Volatility



This exhibit shows that the RP portfolio's volatility is less than half the volatility of the BM portfolio.

D. Sharpe Ratio



As a result, this exhibit shows that the RP portfolio's Sharpe ratio is nearly twice that of the BM portfolio.

[LOS 4.c] Interpret metrics and visuals reported in a backtest of an investment strategy

1.

Metrics

In addition to the familiar Sharpe ratio and the Sortino ratio (which uses target semideviation rather than standard deviation), a number of other measures are commonly used to quantify investment performance such as:

- *Value at Risk*
- *Conditional VaR*
- *Maximum Drawdown*

1.1. Value at Risk (VaR)

VaR is a measure of the minimum amount of value losses expected for a given period of time at a particular level of probability. It is one way to quantify the amount of risk present in the **left tail** of a portfolio's return distribution.

Unfortunately, VaR is sensitive to our assumptions about the shape of the distribution (for example, fat tails versus normal tails).

[LOS 4.c] Interpret metrics and visuals reported in a backtest of an investment strategy

1.

Metrics

1.2. Conditional VaR

Conditional VaR (CVaR) is an extension of VaR that quantifies the weighted average of all returns that exceed the specified value of loss.

If we have a series of historical returns, CVaR is calculated as the average loss once the VaR cutoff is exceeded.

Example: If we select a significance level of $\alpha = 5\%$, the CVaR is the expected value of the return within that lowest 5% of outcomes.

1.3. Maximum drawdown

Maximum drawdown is a commonly used measure of downside risk that represents the *greatest amount of loss* that an asset has experienced *from peak to trough* in the past.

To calculate maximum drawdown, we begin with the asset's maximum cumulative return, and then subtract from the lowest cumulative return that occurs after that point.

Commodity trading advisors (CTAs) and hedge funds often use maximum drawdown.

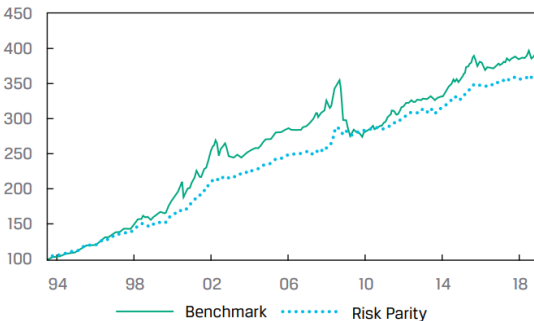
[LOS 4.c] Interpret metrics and visuals reported in a backtest of an investment strategy

2.

Visuals

When evaluating an investment strategy, it can be useful to examine the **long-term cumulative return** performance of an investment strategy by plotting it using a logarithmic scale.

Cumulative Wealth



By examining such a graph, we can often immediately recognize structural breaks and performance decay, and also evaluate downside risk.

[LOS 4.d] Identify problems in a backtest of an investment strategy

1. Survivorship bias

Survivorship bias occurs when, at a particular point in time, we consider only those firms or stocks that have remained in existence (i.e., survived) to that time. It is the most common error that investors make in backtesting.

We can illustrate this issue as the following:

Prior to year 1:



Year 1:



Only surviving companies are added to the index.



Ceased to exist



Survived

[LOS 4.d] Identify problems in a backtest of an investment strategy

1. Survivorship bias

What is the solution to the survivorship bias

Using **point-in-time data**, which represents the specific data that was available at a particular point in history.

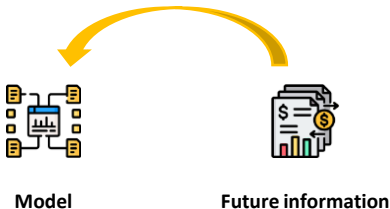
Example: We might consider when analyzing survivorship bias would be the low-instability anomaly, which suggests that over the long term, **low-volatility** stocks will generally **outperform high-volatility** stocks.

- If we correctly use point-in-time data, we would conclude that low-volatility stocks do indeed outperform high volatility stocks from 1989 through 2019.
- If we base our analysis only on those stocks that exist today, without considering the stocks that have disappeared over the decades, we will come to the opposite conclusion.

[LOS 4.d] Identify problems in a backtest of an investment strategy

2. Look-ahead bias

Look-ahead bias is another common error that arises when an investor makes use of data that would not have been available at the time an investment decision is made.



Look-ahead can come about due to **reporting lag**, which refers to a situation where data describing a period is often available *only after* the period ends (and is often subject to **revision**).

[LOS 4.d] Identify problems in a backtest of an investment strategy

2. Look-ahead bias

? What is the solution to the look-ahead bias

Making some adjustments to account for look-ahead bias: *adjust the data by adding a lag assumption of one, two, or three months.*

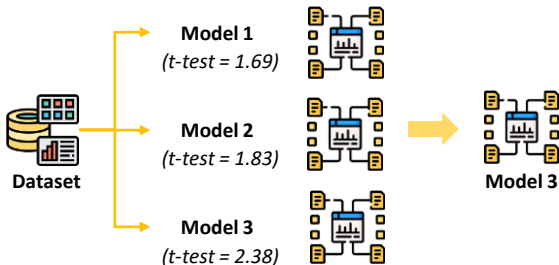
Example: If a firm's financial quarter ends on December 31, 2023, we should not expect to have that company's results on that date. Our first approach would be to make some assumptions about reporting lag.

- If we make the assumption that earnings data for all companies is available a month later on January 31, 2024, we would encounter look-ahead bias with firms that took longer than a month to report earnings.
- If we used a three-month lag and assumed that earnings were available on March 31, 2024, we would be introducing the problem of stale data into our analysis, and our backtest would be overly conservative.

Again, using **point-in-time data** is *preferable* to having to make these kinds of assumptions and adjustments.

[LOS 4.d] Identify problems in a backtest of an investment strategy**3. Data snooping**

Data snooping (p-hacking) is the act of testing multiple investment models and selecting the best-performing one based on statistical results rather than solid theoretical foundations, often leading to a higher risk of *false-positive* outcomes.



Data snooping can come in various forms. A portfolio manager should be wary of strategies that may have been developed using **data snooping** to select the model with the *highest t -statistic* and *lowest p -value*.

[LOS 4.d] Identify problems in a backtest of an investment strategy

3. Data snooping

? *What is the solution to the data snooping*

Using a **higher-than-normal critical t-statistic** (e.g., 3.0) as a benchmark for declaring a variable to be significant.

Using **cross-validation**, a technique that involves testing a hypothesis on a different set of data than the one that was initially used to form the inference or test the hypothesis. To perform **cross-validation**, we divide our data into “training data” versus “validation data”.

Rolling-window backtesting is actually a type of cross-validation, without the random component: in-sample periods (i.e., past periods) are used to tune a model, which is then applied to data from out-of-sample (i.e., later) periods.

[LOS 4.e] Evaluate and interpret a historical scenario analysis

1. Historical scenario analysis

Historical scenario analysis (historical stress testing) is a form of backtesting where we investigate the risk and return of an investment strategy during various regimes, while taking into consideration the various structural breaks that are present.

Example: Historical scenario analysis tests how model handles a move from economic expansions to recessions and from low-volatility to high-volatility environments (and vice versa).

01 Recession versus expansions

The beginning and end of such a downturn are defined retroactively; it is common for the beginning or end of a recession to not be formally identified by economic research organizations until a year or more later.

02 Low- and high-volatility periods

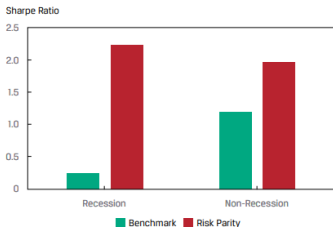
We might define a high-volatility period as one where the VIX is above its five-year moving average, and vice versa.

[LOS 4.e] Evaluate and interpret a historical scenario analysis

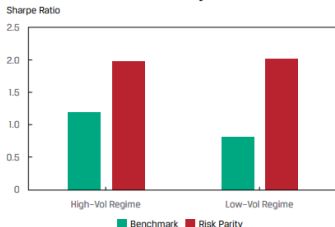
1. Historical scenario analysis

We can examine the **benchmark** and **risk parity factor portfolios** with respect to these two regimes—*recession versus expansion* and *high volatility versus low volatility*.

A. Economic Scenarios



B. Market Volatility Scenarios



In terms of the **Sharpe ratio**

- The RP strategy is quite robust to recession.
- The BM strategy struggles in recessions.

In terms of the **Sharpe ratio**

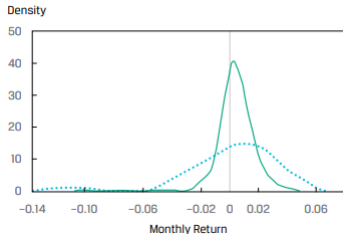
- The BM strategy's performance is slightly worse in low-volatility regimes than in high-volatility regimes.
- The RP strategy performs equally well in both volatility environments.

[LOS 4.e] Evaluate and interpret a historical scenario analysis

1. Historical scenario analysis

In addition to the **Sharpe ratio**, a **probability density plot** can reveal additional information about the sensitivity of the return distributions of these investment strategies - for example, during recession versus non-recession periods.

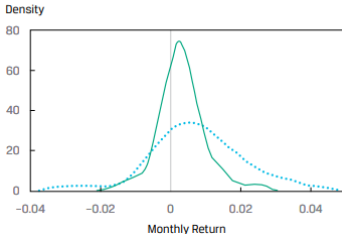
A. Benchmark



— Recession Regime Non-Recession Regime

The BM strategy suffers from negative skewness and excess kurtosis (i.e., fat tails to the left), regardless of the recession regime, but its average return is clearly lower in a recession environment.

B. Risk Parity



The RP strategy also has a lower average return in the recession regime, but its volatility and kurtosis are both also much lower compared with those of the BM strategy.

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

1. Simulation analysis

Recall: The problems of backtesting

Backtesting implicitly assumes that *the past is likely to repeat itself*.

→ This assumption does not fully account for the dynamic nature of financial markets, which may include extreme upside and downside risks that have never occurred before.

We now explore **how simulation can provide a more complete picture**

? What is the role of simulation analysis

- A particularly useful **application of simulation** is to try to *evaluate an investment strategy's downside risk*.
- Simulation can also help us to *validate the results of backtesting*.

? How many types of simulation analysis are there

There are *two* basic types of simulation:

- Historical simulation, and
- Monte Carlo simulation

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

2.

Historical simulation

In a **historical simulation**, we select past returns at random from a long historical period, without regard to the order of occurrence.

→ This methodology is similar to **rolling-window backtesting**, except that we abandon the time sequence in which the returns actually occurred.

? *What is the strength of historical simulation*

Because it relaxes a key restriction by randomly changing the sequencing of historical periods from which factor returns are drawn, as a result, **historical simulation** is essentially a **non-deterministic rolling-window backtest**.

Historical simulation is widely used in *investment management*, particularly by banks for market risk analysis.

? *What is the downside of historical simulation*

We only have actual past data to draw from, and the past only occurred in one particular way.

We need to consider another simulation, in particular, **Monte Carlo simulation**.

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

3. Monte Carlo simulation

Monte Carlo simulation overcomes the downside of **historical simulation** by assigning a statistical distribution to each relevant variable, and then drawing random observations from this distribution.

What is the strength of Monte Carlo simulation

It is very *flexible*, allowing a variety of distributions and actual characteristic such as *non-normality*, *tail dependence*, *fat tails*,... in our key variables, rather than forcing a normal distribution to the data or limiting ourselves to actual historical results.

What is the downside of Monte Carlo simulation

It is complex that necessitates significant computing power.

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

4. Steps of simulation analysis

There are 8 steps typically used to perform a simulation analysis:

01 Select the target variable that we want to analyze

Usually, we are trying to evaluate the *return* of a particular strategy, as well as its *return distribution*.

02 Determine the key decision variables

These include the *returns* of the underlying assets, and the portfolio *weights* of each asset as indicated by the strategy.

03 Select the number of simulation trials (“N”) to run

- Choosing an optimal number of trials is complex; however, a number in the range of 1,000 to 10,000 is typically used.
- Generally, the greater the number of trials, the more stable the results will be.

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

4. Steps of simulation analysis

04 Specify a distribution for each key variable

Here, historical simulation and Monte Carlo simulations differ:

- Historical simulation: we randomly select samples from historical data.
- Monte Carlo simulation: we choose an appropriate statistical distribution (e.g., normal, uniform, lognormal) for each key variable and can also take tail dependence and correlations between variables into account.

05 Draw “N” random numbers for each key variable

A random number generator is used to select these values.

06 Compute the values of the target variable

- Using the values of the key decision variables determined in the previous step, we calculate the corresponding value of the target variable (such as portfolio return).
- These results are compiled for use in the final step.

[LOS 4.f] Contrast Monte Carlo and historical simulation approaches

4. Steps of simulation analysis

07 Repeat random number draws and calculate target variable

Steps 5 and 6 are repeated for “N” iterations.

08 Calculate metrics

- Using the “N” simulated values of the target variable, we compute measures such as average return, variance, Sharpe ratio, etc.
- We can also calculate various downside risk metrics, such as maximum drawdown and CVaR.

[LOS 4.g] Explain inputs and decisions in simulation and interpret a simulation

1.

Historical simulation process

*First, we need decide whether to sample from the historical returns **with replacement** or **without replacement**.*

Random sampling **with replacement**, also known as **bootstrapping**, is often used in investment research because the number of simulations needed is often larger than the size of the historical dataset.

Using the factor allocation strategies (BM and RP) for the eight factor portfolios as an example, we can perform a historical simulation as follows:

01

The target variables are the returns for the BM and RP multifactor portfolios.

02

The key decision variables are the returns of the eight underlying factor-based portfolios.

03

The simulation will be performed for $N = 1,000$ trials.

04

The historical simulation will be implemented using bootstrapped sampling.

[LOS 4.g] Explain inputs and decisions in simulation and interpret a simulation

1.

Historical simulation process

05

Iterate 1000 times with replacement to get all factor returns.

06

Weight each factor return to get BM (equal weights) and RP (weights from final month of period) returns.

07

The same simulation process (from Steps 5 to 6) is repeated for all 1,000 trials, generating a collection of 1,000 simulated returns.

08

Calculate performance metrics of interest (Sharpe ratio, CVaR, etc.) and plot the distributions returns.

[LOS 4.g] Explain inputs and decisions in simulation and interpret a simulation

2.

Monte Carlo simulation process

Monte Carlo simulation follows similar steps as historical simulation but requires a distribution description for each key decision variable.

Exploratory data analysis—focusing on moments (i.e., mean, standard deviation, skewness, kurtosis) and tail dependence—is often *crucial* here.

Regression and distribution-fitting techniques are used to estimate the parameters (i.e., mean, standard deviation, skewness, kurtosis) underlying the statistical distributions of the key decision variables.

→ This step is typically called **model calibration**.

Before finalizing our choice of the functional form of the statistical distribution, we need to account for the following considerations:

01

The distribution should reasonably describe the key empirical patterns of the underlying data.

For example, asset returns roughly follow a bell curve pattern; therefore, the normal distribution and Student's t-distribution are often used as first-cut approximations.

[LOS 4.g] Explain inputs and decisions in simulation and interpret a simulation

2.

Monte Carlo simulation process

02

It is equally critical to account for the correlations between multiple key decision variables.

The returns from multiple factors are clearly correlated; therefore, we need to specify a multivariate distribution rather than modeling each factor or asset on a standalone basis.

03

The complexity of the functional form and number of parameters that determine the functional form are important.

We should recognize the phenomenon called the **bias-variance trade-off** that introduced in earlier readings on machine learning and big data projects.

[LOS 4.h] Demonstrate the use of sensitivity analysis

1. Sensitivity analysis

Sensitivity analysis can be used to gain a better understanding of the risks and returns of an investment strategy, in particular, how our target variable (typically portfolio return) varies due to changes in one of the various input variables.

Sensitivity analysis

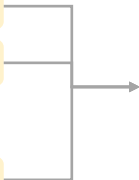
Input variable 1

Input variable 2

:
:

Input variable n

Target variable
(typically portfolio return)



[LOS 4.h] Demonstrate the use of sensitivity analysis

1. Sensitivity analysis

? What is the role of sensitivity analysis

Recall: The result of a Monte Carlo simulation depends heavily on whether we have accurately specified the distributions of underlying variables.

To quantify the risk of misspecification, we can conduct a **sensitivity analysis** by calibrating *a different distribution* to our factor return data and then running the simulation again.

? What is the process of sensitivity analysis

It is *similar* to the process for performing a Monte Carlo simulation, except for fitting the data to a **multivariate skewed t-distribution** rather than to a multivariate normal distribution.

Generating simulations using this new distribution

Evaluating the strategy risk and return indicated by the results

Comparing the results of backtesting and simulation, using measures such as *Sharpe ratio* or *CVaR*



MODULE 5: ECONOMICS AND INVESTMENT MARKETS

LEARNING OUTCOMES

- 5.a.** Explain the notion that to affect market values, economic factors must affect one or more of the following: 1) default-free interest rates across maturities, 2) the timing and/or magnitude of expected cash flows, and 3) risk premiums.
- 5.b.** Explain the role of expectations and changes in expectations in market valuation.
- 5.c.** Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real short-term interest rates.
- 5.d.** Explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities.
- 5.e.** Describe the factors that affect yield spreads between non-inflationadjusted and inflation-indexed bonds.
- 5.f.** Explain how the phase of the business cycle affects credit spreads and the performance of credit-sensitive fixed-income instruments.
- 5.g.** Explain how the characteristics of the markets for a company's products affect the company's credit quality.

LEARNING OUTCOMES

- 5.h.** Explain the relationship between the consumption-hedging properties of equity and the equity risk premium.
- 5.i.** Explain how the phase of the business cycle affects short-term and long-term earnings growth expectations.
- 5.j.** Describe cyclical effects on valuation multiple
- 5.k.** Describe the economic factors affecting investment in commercial real estate.

[LOS 5.a] Explain the notion that to affect market values, economic factors must affect one or more of the following: 1) default-free interest rates across maturities, 2) the timing and/or magnitude of expected cash flows, and 3) risk premiums.

The value of any asset can be computed as the present value of its expected future cash flows discounted at an appropriate risk-adjusted discount rate. The **more uncertain** the cash flows, the **higher** the discount rate.



Components of the discount rate are:

The real risk-free rate (R)

Expected inflation (π)

A risk premium reflecting uncertainty about the cash flow (RP)

The value of an asset will change if either:

- The cash flow forecasts change or;
- Any of the components of the discount rate changes. Risk premiums not only vary across assets (and asset classes), but also vary with changes in investors' perception of risk.

[LOS 5.b] *Explain* the role of expectations and changes in expectations in market valuation.

As market participants receive new information, the timing and amounts of expected future cash flows are revised and valuations change as a result.

The impact of new information will depend on its effect on current expectations.

Example:

An earnings report of 53% growth in earnings may have a positive or negative effect on the firm's value, depending on whether expectations were for slower or more rapid growth.

[LOS 5.c] Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real shortterm interest rates.

1.

Real risk-free rate of return

Even in a world of no inflation, a default-free bond has to compensate an investor for forgoing their current consumption.

The investor evaluates the disutility of forgoing **current consumption** relative to the utility of obtaining **future consumption**.

Inter-temporal rate of substitution

$$m_t = \frac{\text{marginal utility of consuming 1 unit in the future at time } t}{\text{marginal utility of current consumption of 1 unit at time } 0} = \frac{u_t}{u_0}$$

For a given quantity of consumption, investors always prefer current consumption over future consumption ($u_0 > u_t$) and $m_t < 1$ as a result.

The current price (P_0) of a zero-coupon, inflation-indexed, risk-free bond that will pay \$1 at time 1 can be expressed as: $P_0 = E(m_1)$

Real risk free rate of return

$$R = \frac{1 - P_0}{P_0} = \frac{1}{E(m_1)} - 1$$

[LOS 5.c] Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real shortterm interest rates.

1. Real risk-free rate of return

Some key points to keep in mind:

- 1.** The higher the utility investors attach for current consumption relative to future consumption → The higher the real rate
- 2.** Diminishing marginal utility of wealth means that an investor's marginal utility of consumption declines as wealth increases.
→ Marginal utility of consumption is higher during periods of scarcity, such as during economic contractions.
- 3.** When investor expectations about the economy change to better:
→ Expectation of higher incomes → increase in current consumption and a reduction in savings.
→ Investors will derive greater utility from current consumption relative to future consumption.
- 4.** Investors increase their savings rate when expected returns are high or when uncertainty about their future income increases.

[LOS 5.c] Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real shortterm interest rates.

2.

Risky Cash Flows and Risk Premiums

- If the underlying cash flows are uncertain, investors demand a risk premium.
- The investor's expected marginal utility of a payoff is ***inversely related*** to the level of uncertainty of the payoff.

Risk aversion

Investors experience a larger loss of utility for a loss in wealth as compared to a gain in utility for an equivalent gain in wealth

- An investor's absolute risk aversion **declines with their wealth**: wealthier investors are *less risk-averse and more willing to take risk* relative to their poorer counterparts.
- However, the **marginal utility** of holding risky assets **declines** as an investor holds more risky assets in her portfolio.

When the markets are in equilibrium, wealthy and poorer investors would have the **same willingness** to hold risky assets.

[LOS 5.c] Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real shortterm interest rates.

2.

Risky Cash Flows and Risk Premiums

Price of risky bond

$$P_0 = \frac{E(P_1)}{(1+R)} + \text{cov}(P_1, m_1)$$

Where:

R = the real risk-free rate

$\text{cov}(P_1, m_1)$ = covariance between the expected future price of the bond and the investor's inter-temporal rate of substitution → **risk premium**. The uncertainty about the sale price gives rise to a risk premium.

During good economic times, both investors' labor incomes and most risky asset values are **high** but the marginal utility of future consumption is **lower**.

Negative covariance, or **negative risk premium**

Reduces the value of P_0
→ higher expected return

→ The price of the bond will be lower than the expected sale price discounted at the real risk free rate.

[LOS 5.c] Explain the relationship between the long-term growth rate of the economy, the volatility of the growth rate, and the average level of real shortterm interest rates.

3.

GDP Growth Rates

GDP growth is forecasted to be high

01



Utility consumption of in the future will be low (m_t fall)

02



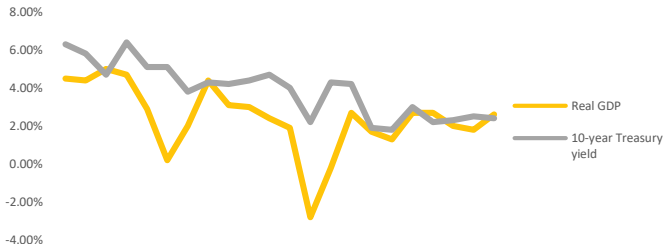
Investor save less, increasing interest rates

03



Real interest rates will be **positively correlated** with real GDP growth rates and with expected volatility in GDP growth due to higher risk premium.

Real GDP growth vs US 10-year yield (1996-2016)



[LOS 5.d] Explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities.

Short-term interest rate

$$\begin{array}{c} \text{nominal} \\ \text{interest rate} \\ (r_{\text{short}}) \end{array} = \begin{array}{c} \text{real} \\ \text{risk-free rate (R)} \end{array} + \begin{array}{c} \text{expected} \\ \text{inflation } (\pi) \end{array}$$

Long-term interest rate

Because actual inflation is uncertain → gives rise to additional risk premium for **the uncertainty about actual inflation (θ)**. This risk premium is higher for longer maturity bonds while negligible for short-term risk-free securities.

$$\begin{array}{c} \text{nominal} \\ \text{interest rate} \\ (r_{\text{long}}) \end{array} = \begin{array}{c} \text{real} \\ \text{risk-free rate} \\ (R) \end{array} + \begin{array}{c} \text{expected} \\ \text{inflation } (\pi) \end{array} + \begin{array}{c} \text{premium for} \\ \text{uncertainty} \\ \text{inflation } (\theta) \end{array}$$

[LOS 5.d] Explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities.

1.

Taylor Rule


Central bank
setting policy
rates to:

Achieve the maximum
sustainable level of
employment

Maintain price stability

The **Taylor rule** links the central bank's policy rate to economic conditions (employment level and inflation).

Taylor rule

$$r = R_n + \pi + 0.5 (\pi - \pi^*) + 0.5 (y - y^*)$$

r = central bank policy rate implied by the Taylor rule

R_n = neutral real policy interest rate

π = current inflation rate

π^* = central bank's target inflation rate

y = log of current level of output

y^* = log of central bank's target (sustainable) output

Central banks can moderate the business cycle by making changes to the policy rate or can magnify the cycle by not responding appropriately to changing economic conditions

[LOS 5.d] Explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities.

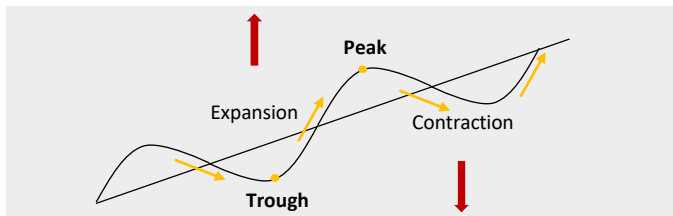
2.

Business Cycle and Slope of the Yield Curve

**Negatively sloped
yield curve**

Investor's expectation:

- *Decreasing inflation*
- *Decreasing GDP growth at later expansion stage.*



Policy rates tend to be low

Higher longer-term rates

**Positively sloped
yield curve**

Investors' expectation:

- *Improving future GDP growth*
- *Increasing inflation as the economy comes out of recession*

[LOS 5.d] Explain how the phase of the business cycle affects policy and short-term interest rates, the slope of the term structure of interest rates, and the relative performance of bonds of differing maturities.

2.

Business Cycle and Slope of the Yield Curve

Term spread

The difference between the yield on a longer-term bond yield and the yield on a short-term bond.

Recall: *the risk premium for uncertainty in inflation (θ) is higher for longer maturity bonds.*



Normal term spread is positive so the yield curve is upward sloping and positive term spreads can be attributed to increasing θ for longer periods.

[LOS 5.e] Describe the factors that affect yield spreads between non-inflationadjusted and inflation-indexed bonds.

**Break-even
inflation
rate (BEI)**

The difference between the yield of a non-inflation-indexed risk-free bond and the yield of an inflation-indexed risk-free bond of the same maturity.

Break-even inflation rate

$$\text{BEI} = \text{yield on non-inflation-indexed bond} - \text{yield on inflation-indexed bond}$$

Recall: for longer maturity bonds, the nominal rate is composed of the real rate, expected inflation, and a risk premium for inflation uncertainty.

Break-even
inflation rate (BEI)



Expected inflation (π)

Risk premium for
uncertainty about
actual inflation (θ)

[LOS 5.f] Explain how the phase of the business cycle affects credit spreads and the performance of credit-sensitive fixed-income instruments.

The required rate of return for bonds with credit risk includes **credit risk premium (credit spread)**.

Credit spread

The difference in yield between a credit risky bond and a default-free bond of the same maturity.

Return
rate for
credit
risky
bonds

=

real
risk-free
rate (R)

+

expected
inflation
(π)

+

premium for
uncertainty
inflation (θ)

+

credit
spread
(γ)

*Economic
downturn*

Credit spreads tend to rise.

→ higher rated bonds will outperform lower rated bonds on a relative basis.

*Economic
expansion*

Credit spreads narrows.

→ credit risky bonds will outperform default-free bonds.



[LOS 5.g] Explain how the characteristics of the markets for a company's products affect the company's credit quality

Differences in credit spreads by sectors are primarily due to differences in **industry** products and services and the financial leverage of the firms in the industry.



**Consumer cyclical sector
spreads**



**Consumer noncyclical
sector spreads**

Spreads for issuers in the **consumer cyclical sector** increase significantly during economic downturns compared to spreads for issuers in the **consumer noncyclical sector**.

[LOS 5.h] Explain the relationship between the consumption-hedging properties of equity and the equity risk premium.

The discount rate used to value equity securities includes an additional risk premium, **the equity risk premium**. This risk premium is in addition to the risk premium on credit risky bonds because equity is more risky than debt.

$$\text{Discount rate for equity} = R + \pi + \theta + \gamma + \text{additional risk premium } (\kappa)$$

$$\lambda = \text{equity risk premium} \\ = \gamma + \kappa$$

Assets that provide a higher payoff during economic downturns are more highly valued because of the **consumption hedging property** of the asset
→ Reduces the risk premium on an asset.

Equity prices: cyclical, with higher values during economic expansions when the marginal utility of consumption is lower.

→ Equity investments are not the most effective hedge against bad consumption outcomes.

→ Equity risk premium is positive.

[LOS 5.i] : *Explain* how the phase of the business cycle affects short-term and long-term earnings growth expectations.

Corporate earnings may be related to the business cycle.

Cyclical industries

- Relatively more sensitive to the phase of the business cycle
- Revenues and earnings that rise and fall with the rate of economic growth.

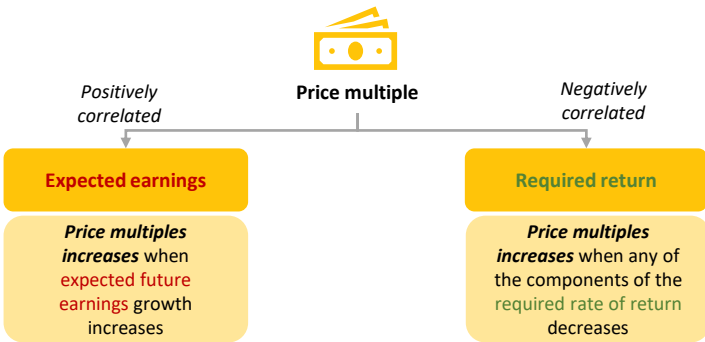


Defensive / non-cyclical industries

- Relatively immune to fluctuations in economic activity
- Earnings tend to be relatively stable throughout the business cycle.

[LOS 5.j]: Describe cyclical effects on valuation multiples.

It is inappropriate to judge the multiples (such as P/E and P/B) in a historical context **only** in determining the relative values of companies, of sectors, or of the overall market.

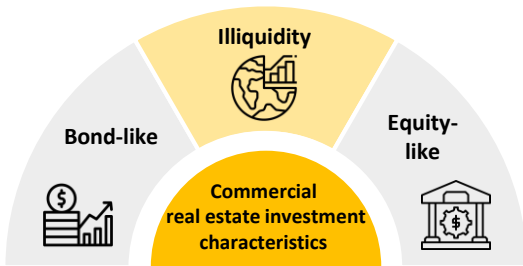


→ The equity risk premium ↓ during economic expansions and ↑ during recessions.

**Shiller's
CAPE**

CAPE ratio reduces the volatility of unadjusted P/E ratios by using real prices in the numerator and a 10-year moving average of real earnings in the denominator.

[LOS 5.k]: Describe the economic factors affecting investment in commercial real estate.



The steady rental income stream is similar to ***bond-like securities***.

→ Credit quality of tenants affects the value of real estate.

Real estate as an asset class is characterized by ***illiquidity***; it could take years to exit a real estate investment at its fair value.

Uncertainty about the value of the property at the end of the lease term gives an ***equity-like character***.

[LOS 5.k]: *Describe* the economic factors affecting investment in commercial real estate.

Valuation

When estimating the value of real estate investment, the discount rate includes an additional risk premium for the lack of liquidity.

$$\text{Discount rate for R.E.} = R + \pi + \theta + \gamma + \kappa + \text{Illiquidity premium } (\phi)$$

While rental income from commercial properties seems to be more or less steady across business cycles, commercial property values tend to be very **cyclical**

Correlation of commercial property values with those of other asset classes (e.g., equities) tends to be positive

Real estate provides a poor hedge against bad consumption outcome

Risk premium will be relatively **high** and often close to the risk premium required for equity investment



MODULE 6: ANALYSIS OF ACTIVE PORTFOLIO MANAGEMENT

LEARNING OUTCOMES

- 6.a.** Describe how value added by active management is measured
- 6.b.** Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio
- 6.c.** Describe and interpret the fundamental law of active portfolio management, including its component terms—transfer coefficient, information coefficient, breadth, and active risk (aggressiveness)
- 6.d.** Explain how the information ratio may be useful in investment manager selection and choosing the level of active portfolio risk
- 6.e.** Compare active management strategies, including market timing and security selection, and evaluate strategy changes in terms of the fundamental law of active management
- 6.f.** Describe the practical strengths and limitations of the fundamental law of active management

[LOS 6.a] Describe how value added by active management is measured

1.

Choice of benchmark

Active management seeks to *add value* by *outperforming* a passively managed *benchmark portfolio*.

A benchmark or passive portfolio should have a number of qualities to serve as a relevant comparison for active management:

01

The benchmark is representative of the assets from which the investor will select.

02

Positions in the benchmark portfolio can actually be replicated at low cost.

03

Benchmark weights are verifiable *ex ante*, and return data are timely *ex post*.

An available *security market index* is often used as the *benchmark portfolio*. The *most common market indexes* weight the individual assets by their *market capitalization*.

[LOS 6.a] Describe how value added by active management is measured

1.

Choice of benchmark

The return on the benchmark portfolio, R_B , is based on the returns to the individual securities and the weights of each security in the portfolio:

$$R_B = \sum_{j=1}^N w_{B,j} \cdot R_j$$

where:

R_j is the return on security j

$w_{B,j}$ is the benchmark weight of security j

N is the number of securities

Similarly, the return on an actively managed portfolio, R_P , is a function of the weights of the securities, j , held in the portfolio, $w_{P,j}$, and the returns to the individual securities:

$$R_P = \sum_{j=1}^N w_{P,j} \cdot R_j$$

Note that for simplicity, the same notation, N , is used in the summation in the expression for the managed portfolio return and the benchmark return, although fewer or more securities may be in the managed portfolio than in the benchmark.

[LOS 6.a] Describe how value added by active management is measured

2.

Measuring value added

The **value added** or “**active return**” of an actively managed portfolio is typically calculated as the *simple difference* between the *return on that portfolio* and the *return on the benchmark portfolio*.

$$R_A = R_P - R_B$$

Assumption: the systematic risk (β) of the active portfolio (β_P) is the *same* as the systematic risk of the benchmark portfolio (β_B).

If β_P is *different* than β_B , **active return** is computed as the difference in *risk-adjusted returns* and is known as **alpha** (α_P).

$$\alpha_P = R_P - \beta_P \cdot R_B$$

Active return can be either positive or negative, and it can be measured *ex-ante* (i.e., based on expectations) or *ex-post* (“after the fact”).

[LOS 6.a] Describe how value added by active management is measured

2.

Measuring value added

We will provide a more detailed illustration of active return:

$$R_A = R_P - R_B$$

$$\begin{aligned} R_A &= \sum_{j=1}^N w_{P,j} \cdot R_j - \sum_{j=1}^N w_{B,j} \cdot R_j \\ &= \sum_{j=1}^N (w_{P,j} - w_{B,j}) \cdot R_j \end{aligned}$$

where $\Delta w_j = w_{P,j} - w_{B,j}$ is the **active weights** of the managed portfolio

Because $\sum w_{P,j} = \sum w_{B,j} = 1$, we must have $\sum \Delta w_j = 0$ (the sum of the active weights is always **zero**).

Then, we can also write the value added as the sum product of active weights and active security returns:

$$R_A = \sum_{j=1}^N \Delta w_j \cdot R_{Aj}$$

where $R_{Aj} = R_j - R_B$ is the **active security return** of security j

[LOS 6.a] Describe how value added by active management is measured

2.

Measuring value added

Positive value added

Securities j that have $R_j > R_B$ are overweighted and securities j that have $R_j < R_B$ are underweighted.

Negative value added

Securities j that have $R_j < R_B$ are overweighted and securities j that have $R_j > R_B$ are underweighted.

Example: Active return

The following information is available for an actively managed portfolio and its benchmark.

Calculate the ex-ante active return.

Security (i)	Portfolio weight (w_{Pi})	Benchmark weight (w_{Bi})	Expected return $E(R_i)$
A	22%	25%	12%
B	20%	25%	-6%
C	21%	25%	4%
D	37%	25%	19%

[LOS 6.a] Describe how value added by active management is measured

2. Measuring value added

Example: Active return

Answer:

Recall: Active return = $E(R_P) - E(R_B)$

Security (i)	Portfolio weight (w_{Pi})	Benchmark weight (w_{Bi})	Expected return $E(R_i)$	Active weight (Δw_i)
A	22%	25%	12%	-3%
B	20%	25%	-6%	-5%
C	21%	25%	4%	-4%
D	37%	25%	19%	12%

$$E(R_P) = \sum w_{Pi} \cdot E(R_i) = 22\% \cdot 12\% + 20\% \cdot -6\% + 21\% \cdot 4\% + 37\% \cdot 19\% = 9.31\%$$

$$E(R_B) = \sum w_{Bi} \cdot E(R_i) = 25\% \cdot 12\% + 25\% \cdot -6\% + 25\% \cdot 4\% + 25\% \cdot 19\% = 7.25\%$$

$$\rightarrow E(R_A) = E(R_P) - E(R_B) = 9.31\% - 7.25\% = 2.06\%$$

Alternatively:

$$E(R_A) = \sum \Delta w_i \cdot E(R_i) = -3\% \cdot 12\% + -5\% \cdot -6\% + -4\% \cdot 4\% + 12\% \cdot 19\% = 2.06\%$$

[LOS 6.a] Describe how value added by active management is measured

3.

Decomposition of active return

Active return can be decomposed into two parts:

$$R_A = \sum \Delta w_j \cdot R_{B,j} + \sum w_{P,j} \cdot R_{A,j}$$

01

Asset allocation return
(from deviations of asset class portfolio weights from benchmark weights)

02

Security selection return
(from active returns within asset classes)

where:

$\sum \Delta w_j \cdot R_{B,j}$ = return from asset allocation

$\sum w_{P,j} \cdot R_{A,j}$ = return from security selection

$R_{A,j} = R_{P,j} - R_{B,j}$ = expected active return within asset classes

Example: Consider an active portfolio manager with a benchmark portfolio composed of stocks and bonds.

- **Asset allocation return:** Overweight stocks (bonds) and underweight bonds (stocks).
- **Security selection return:** Overweight a specific stock (bond) within the allocation to stocks (bonds) (and correspondingly underweight some other stocks (bonds)).

[LOS 6.a] Describe how value added by active management is measured

3.

Decomposition of active return

Example: Active return

Optoma Fund invests in three asset classes: U.S. equities, U.S. bonds, and international equities. The asset allocation weights of Optoma and the expected performance of each asset class and the benchmark are shown in the following table.

Calculate the expected active return from asset allocation and security selection. Then calculate the total expected active return.

Asset class (i)	Portfolio weight (w_{Pi})	Benchmark weight (w_{Bi})	Portfolio return $E(R_{Pi})$	Benchmark return $E(R_{Bi})$
U.S. equities (E)	45%	40%	11%	12%
U.S. bonds (B)	30%	30%	6%	5%
International equities (E)	25%	30%	14%	12%

[LOS 6.a] Describe how value added by active management is measured

3.

Decomposition of active return

Example: Active return

Answer:

Recall: We have the expected return $E(R_A) = \sum \Delta w_j \cdot E(R_{B,j}) + \sum w_{P,j} \cdot E(R_{A,j})$
 where $\sum \Delta w_j \cdot E(R_{B,j})$ is the **return from asset allocation (R_{Aa})**.

$\sum w_{P,j} \cdot E(R_{A,j})$ is the **return from security selection (R_{SS})**.

Asset class (i)	(w_{Pi})	(w_{Bi})	$E(R_{Pi})$	$E(R_{Bi})$	R_{Aa}	R_{SS}
	(1)	(2)	(3)	(4)	$((1) - (2)) \times (4)$	$((3) - (4)) \times (1)$
U.S. E	45%	40%	11%	12%	0.60%	-0.45%
U.S. B	30%	30%	6%	5%	0.00%	0.30%
International E	25%	30%	14%	12%	-0.60%	0.50%
Total					0.00%	0.35%

Thus, the return from asset allocation is 0%, and the return from security selection is 0.35%, resulting in an expected active return of 0.35%.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Sharpe Ratio

The **Sharpe ratio** is used to compare the portfolio return in excess of a riskless rate with the volatility of the portfolio return.

It provides a measure of how much the investor is receiving in excess of a riskless rate for assuming the risk of the portfolio.

The Sharpe ratio (SR) is calculated as excess return per unit of risk (standard deviation):

$$SR = \frac{R_P - R_F}{\sigma_P}$$

where:

R_P is the portfolio return

R_F is the risk free rate

σ_P is the standard deviation of the portfolio return

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Sharpe Ratio

An **important attribute of the Sharpe ratio** is that it is *unaffected by the addition of cash or leverage* in the portfolio.

In deed, considering a combined portfolio with a weight of w_P on the actively managed portfolio and a weight of $(1 - w_P)$ on risk-free cash.

→ The return on the combined portfolio is $R_C = w_P R_P + (1 - w_P) R_F$ and the volatility of the combined portfolio is just $\sigma_C = w_P \sigma_P$ because the $(1 - w_P) R_F$ portion is risk free.

Now, we have:

$$SR_C = \frac{R_C - R_F}{\sigma_C} = \frac{w_P R_P + (1 - w_P) R_F - R_F}{w_P \sigma_P} = \frac{w_P R_P - w_P R_F}{w_P \sigma_P} = \frac{R_P - R_F}{\sigma_P} = SR_P$$

- If the expected volatility of the risky asset portfolio is higher than the investor prefers, the volatility can be reduced by holding more cash and less of the risky portfolio.
- If the expected volatility of the risky portfolio is lower than the investor allows, the volatility and expected return can be increased by leverage.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

The **information ratio (IR)** is the ratio of the active return to the standard deviation of active returns, which is known as **active risk** or **benchmark tracking risk**.

$$IR = \frac{R_P - R_B}{\sigma(R_P - R_B)} = \frac{R_A}{\sigma_A} = \frac{\text{active return}}{\text{active risk}}$$

where:

R_P is the portfolio return

R_B is the benchmark return

σ(.) is the standard deviation function—in this case, the standard deviation of the excess return of the portfolio (**R_P**) over the return of the benchmark (**R_B**)

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

There are some important points related to the information ratio:

01

A **closet index fund** will have a SR similar to that of the benchmark index, a very low IR, and little active risk.
After fees, the IR of that fund is often negative.

02

A fund with zero systematic risk that uses the risk-free rate as its benchmark would have an IR that is equal to its SR.

It is clear because if $\mathbf{B} = \mathbf{F}$ then we have $\mathbf{R_B} = \mathbf{R_F}$ and $\sigma(\mathbf{R_P} - \mathbf{R_B}) = \sigma(\mathbf{R_P} - \mathbf{R_F}) = \sigma_P$

03

The IR will change with the addition of cash or the use of leverage.

If the investor adds cash to a portfolio of risky assets, the IR for the combined portfolio will generally shrink, because the active return will decrease, while active risk should not change much.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

There are some important points related to the information ratio:

04

The IR of an unconstrained portfolio is unaffected by the aggressiveness of the active weights.

It is clear because if Δw_j are all multiplied by some constant, c
 $\rightarrow R_C = c.R_A$ and $\sigma_C = c.\sigma_A$

05

Investors can select an appropriate amount of active risk by investing a portion of their assets in the active portfolio and the remaining portion in the benchmark.

Example: If the active risk of a fund is 10%, an investor seeking to limit active risk to 6% can do so by investing 60% in the active portfolio and the remaining 40% in the benchmark portfolio.

06

If we combine an actively managed portfolio with an allocation to the benchmark portfolio, the resulting blended portfolio will have the same IR as the original actively managed portfolio.

Example: If the active risk of a fund is 5.0%, combining that fund in an 80/20 mix with the benchmark portfolio will result in an active risk of the combined portfolio of $0.80(5.0\%) = 4.0\%$, with a proportional reduction in the active return.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

For an unconstrained active portfolio, the optimal amount of active risk is the level of active risk that **maximizes the portfolio's Sharpe ratio**.

This optimal active risk can be calculated as:

$$\sigma_A^* = \frac{IR}{SR_B} \sigma_B$$

The Sharpe ratio of a portfolio with optimal level of active risk can be calculated as:

$$SR_P = \sqrt{SR_B^2 + IR^2}$$

Furthermore, the total risk of the portfolio is given by:

$$\sigma_P^2 = \sigma_B^2 + \sigma_A^2$$

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

Example 1: Optimal active risk

Omega fund has an information ratio of 0.2 and active risk of 9%. The benchmark portfolio has a Sharpe ratio of 0.4 and total risk of 12%. If a portfolio (portfolio P) with an optimal level of active risk has been constructed by combining Omega fund and the benchmark portfolio, calculate:

1. Portfolio P's Sharpe ratio.
2. Portfolio P's excess return (i.e., return above the risk-free rate).
3. The proportion of benchmark and Omega fund in portfolio P.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

Example 1: Optimal active risk

Answer:

1. **Recall: Sharp ratio calculation:** $SR_P = \sqrt{SR_B^2 + IR^2}$

We are given $SR_B = 0.4$ and $IR = 0.2$. Thus, $SR_P = \sqrt{0.4^2 + 0.2^2} = 0.4472$

2. Firstly, **Recall that** portfolio P's excess return = $R_P - R_F$

Given the benchmark Sharpe ratio of 0.4 and a benchmark total risk of 12%,

we have: $0.40 = \frac{R_B - R_F}{\sigma_B} = \frac{R_B - R_F}{0.12} \rightarrow R_B - R_F = 4.8\%$.

Next, we need to calculate $R_P - R_B$

Recall that $R_P - R_B = R_A = IR \cdot \sigma_A$

Note that σ_A is the optimal level of active risk.

Thus, we have $\sigma_A = \frac{IR}{SR_B} \sigma_B = \frac{0.2}{0.4}(12\%) = 6\%$.

$\rightarrow R_P - R_B = IR \cdot \sigma_A = 0.2 \times 6\% = 1.2\%$

Hence, portfolio P's excess return = $R_P - R_F = (R_P - R_B) + (R_B - R_F) = 1.2\% + 4.8\% = 6.0\%$.

3. The optimal level of active risk is 6% and Omega fund has an active risk of 9%.

$\rightarrow$ We can calculate that $6\% / 9\% = 67\%$ of portfolio P's allocation will be to the Omega fund and 33% to the benchmark portfolio.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

Example 2: Optimal active risk

Omega fund has a Sharpe ratio of 0.5. The benchmark portfolio has a Sharpe ratio of 0.4 and total risk of 12%. If a portfolio (portfolio P) with an optimal level of active risk has been constructed by combining Omega fund and the benchmark portfolio, calculate:

1. Portfolio P's Information ratio.
2. The optimal level of active risk.

[LOS 6.b] Calculate and interpret the information ratio (ex post and ex ante) and contrast it to the Sharpe ratio

1.

The Information Ratio

Example 2: Optimal active risk

Answer:

1. *Recall: Sharp ratio calculation:* $SR_P = \sqrt{SR_B^2 + IR^2}$

$$\rightarrow IR = \sqrt{SR_P^2 - SR_B^2} = \sqrt{0.5^2 - 0.4^2} = 0.3$$

Note that the IR above is the IR of Omega fund. But we ***recall that*** if we combine an actively managed portfolio with an allocation to the benchmark portfolio, the resulting blended portfolio will have the same IR as the original actively managed portfolio.

→ Portfolio P's Information ratio = Omega fund's IR = 0.3

2. *Recall: Optimal level of active risk calculation:* $\sigma_A = \frac{IR}{SR_B} \sigma_B$

$$= \frac{0.3}{0.4}(12\%) = 9\%.$$

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms—transfer coefficient, information coefficient, breadth, and active risk (aggressiveness)

1.

The Information Ratio's component terms

There are **three** factors that determine the information ratio:

01

The **information coefficient (IC)**: is a measure of a manager's skill

- Ex-ante IC is the risk-weighted correlation between active returns and forecasted active returns.
- Ex-post IC measures actual correlation between active returns and expected active returns.

02

The **transfer coefficient (TC)**: the correlation between *actual active weights* and *optimal active weights*

The *optimal active weight* for a security is *positively* related to its *expected active return* and *negatively* related to its *expected active risk*.

For an unconstrained active portfolio, the active weights will be equal to the optimal weights and $TC = 1$.

For a constrained portfolio, TC may be less than 1.

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

1.

The Information Ratio's component terms

02

The **transfer coefficient (TC)**: the correlation between *actual active weights* and *optimal active weights*

More precisely, TC is the cross-sectional correlation between the *forecasted active returns* and the *actual weights adjusted for risk*.

$$\begin{aligned}\text{TC} &= \text{CORR}(\mu_i/\sigma_i, \Delta w_i \sigma_i) \\ &= \text{CORR}(\Delta w_i^* \sigma_i, \Delta w_i \sigma_i)\end{aligned}$$

where:

μ_i is the ex-ante active return for security i

Δw_i is the actual active weight for security i

Δw_i^* is the mean-variance optimal active weight for security i

σ_i is the forecasted volatility for security i

03

Breadth (BR): the number of independent active bets taken per year

For example, if a manager takes active positions in 10 securities each month, then $BR = 10 \times 12 = 120$.

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

2.

The Fundamental Law of active portfolio management

The **Grinold rule** allows us to compute the expected active return based on the information coefficient, active risk, and a standardized score.

$$\mu_i = IC \cdot \sigma_i \cdot S_i$$

where:

S_i is the score of security i (Standardized with an assumed variance of 1)

Recall: The expected active portfolio return, is the sum product of active security weights and forecasted active security returns:

$$E(R_A) = \sum \Delta w_i \cdot \mu_i$$

$$E(R_A) = \sum \Delta w_i \cdot IC \cdot \sigma_i \cdot S_i$$

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

2.

The Fundamental Law of active portfolio management

According to the **Grinold rule**, the mean–variance optimal active weights are:

$$\Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \frac{\sigma_A}{IC \cdot \sqrt{BR}}$$



$$\mu_i = IC \cdot \sigma_i \cdot S_i$$

$$\Delta w_i^* = \frac{IC \cdot \sigma_i \cdot S_i}{\sigma_i^2} \frac{\sigma_A}{IC \cdot \sqrt{BR}} = \frac{S_i \sigma_A}{\sigma_i \sqrt{BR}}$$

But from the previous slide, we have the expected active portfolio return:

$$E(R_A) = \sum \Delta w_i \cdot IC \cdot \sigma_i \cdot S_i$$

Thus, note that for an unconstrained portfolio, $TC = 1$ and optimal values are denoted by asterisks (*):

$$E(R_A)^* = \sum \Delta w_i^* \cdot IC \cdot \sigma_i \cdot S_i = IC \cdot \sqrt{BR} \cdot \sigma_A$$

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

2.

The Fundamental Law of active portfolio management

$$E(R_A)^* = IC \cdot \sqrt{BR} \cdot \sigma_A$$

→ The equation above states the **basic fundamental law** of active management.

Using equation above, we can also express the information ratio of the unconstrained optimal portfolio, $E(R_A)^*/\sigma_A$, as the product of just *two* terms: $IR^* = IC \cdot \sqrt{BR}$

For constrained portfolios, the actual active weights (Δw_i) will differ from the optimal active weights (Δw_i^) and the **TC will be less than 1**. In this case, we have:*

$$IR = (TC) \cdot IC \cdot \sqrt{BR}$$

$$E(R_A) = (TC) \cdot IC \cdot \sqrt{BR} \cdot \sigma_A$$

*Note that the transfer coefficients are always less than one (**TC < 1**)*

→ The information ratio must be less than the optimal information ratio (**IR < IR***) and the expected active return must be less than the optimal expected active return **$E(R_A) < E(R_A)^*$** .

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

2.

The Fundamental Law of active portfolio management

Recall: The optimal level of active risk (in an **unconstrained** portfolio) is a function of the information ratio, the Sharpe ratio of the benchmark, and the standard deviation of the benchmark return:

$$\sigma_A^* = \frac{IR}{SR_B} \sigma_B$$

$$\sigma_A^* = \frac{IR}{SR_B} \sigma_B$$

For a **constrained** portfolio, the optimal level of active risk (σ_{CA}^*) is lower than that of an **unconstrained** portfolio and is calculated as:

$$\sigma_{CA}^* = TC \frac{IR^*}{SR_B} \sigma_B$$

where:

IR^* is the information ratio of an unconstrained portfolio

Similarly, the Sharpe ratio of a **constrained** portfolio is lower than that of an unconstrained portfolio and is given by:

$$SR_{PC} = \sqrt{SR_B^2 + TC^2 \cdot IR^{*2}}$$

[LOS 6.c] Describe and interpret the fundamental law of active portfolio management, including its component terms...

3.

Ex-post performance measurement

***Recall:** Realized value added from active management is the ex-post active return that the manager achieves.*

Using the ex-post information coefficient IC_R , the fundamental law can be written as:

$$E(R_A | IC_R) = (TC) \cdot (IC_R) \cdot \sqrt{BR} \cdot \sigma_A$$

The actual return on the active portfolio can be expressed as its conditional expected return and a noise term:

$$R_A = E(R_A | IC_R) + \text{noise}$$

where $E(R_A | IC_R)$ represents the expected value added, given the realized skill of the investor that period, and the noise term is the constraint-induced noise.

The proportion of realized active return variance attributed to variation in the realized information coefficient is TC^2 .

The remaining $(1 - TC^2)$ is attributed to constraint-induced noise.

[LOS 6.d] Explain how the information ratio may be useful in investment manager selection and choosing the level of active portfolio risk

1.

Usefulness of the information ratio

Portfolio theory concludes that investors will choose some combination of the *risk-free asset* and an *optimal risky portfolio (the portfolio with the highest Sharpe ratio)*, with the weights determined by their risk tolerance.

Recall: $SR_P = \sqrt{SR_B^2 + IR^2}$

→ *The Sharpe ratio of an actively managed portfolio is higher than that of the benchmark based on the information ratio.*

Investors will choose the active manager with the *highest information ratio to maximize the Sharpe ratio.*

The actively managed portfolio with the **highest IR** is the optimal (active) portfolio for all investors *regardless of their risk tolerance.*

The IR can be used to determine the expected active return for a given target level of active risk:

$$E(R_A) = IR \cdot \sigma_A$$

[LOS 6.e] Compare active management strategies, including market timing and security selection, and evaluate strategy changes in terms of the fundamental law of active management

1.

Market timing

Market timing is simply a bet on the direction of the market (or a segment of the market).

Example: An active manager may rotate money out of equities and into cash based on an expected decline in stock prices.

For a market timer, the information coefficient is based on the proportion of correct calls.

$$IC = 2(\% \text{ correct}) - 1$$

If the manager is correct 50% the time, the IC will equal zero.

[LOS 6.e] Compare active management strategies, including market timing and security selection, and...

1.

Market timing

Example: Market timer vs. security selector

Darsh Bhansali is a manager with Optimus Capital. Bhansali, a market timer, makes quarterly asset allocation decisions based on his forecast of the direction of the market. Bhansali's forecasts are right 55% of the time. Mike Neal is an equity analyst focusing on technology stocks. Neal, a security selector, typically makes 50 active stock selections annually. Neal has an information coefficient of 0.04. Compute the information ratios of Bhansali and Neal assuming that both managers construct unconstrained portfolios.

Answer:

Because both portfolios are unconstrained, $TC = 1$.

As Bhansali is a market timer, Bhansali's $IC = 2(0.55) - 1 = 0.10$

→ Bhansali's $IR = IC \cdot \sqrt{BR} = 0.10 \times \sqrt{4} = 0.20$

As Neal is a security selector, Neal's $IR = IC \cdot \sqrt{BR} = 0.04 \times \sqrt{50} = 0.28$

[LOS 6.e] Compare active management strategies, including market timing and security selection, and...

2.

Sector rotation

Market timing can also be used to make **sector rotation** decisions (allocating assets into sectors that are expected to outperform).

Consider a two sector market X and Y . Assume the expected sector return and volatility of returns are $E(R_X)$ and σ_X for Sector X , and $E(R_Y)$ and σ_Y for Sector Y .

If the correlation between the returns of sectors X and Y is given by r_{XY} , the active risk of this strategy is the standard deviation of differential returns of the two sectors $(R_X - R_Y)$ and is given by σ_C :

$$\sigma_C = \sqrt{\sigma_X^2 - 2\sigma_X\sigma_Y r_{XY} + \sigma_Y^2}$$

The **annualized** active risk is a function of the number of bets made during the year.

$$\sigma_A = \sigma_C \cdot \sqrt{BR}$$

Example: The active manager makes quarterly bets $\rightarrow BR = 4 \rightarrow \sigma_A = 2\sigma_C$

[LOS 6.e] Compare active management strategies, including market timing and security selection, and...

2.

Sector rotation

Example: Sector rotation

Hwang Soi makes monthly allocation decisions between consumer discretionary and consumer staples based on a proprietary model. The historical correlation between the returns of the two sectors is 0.30 and Soi's bets have been correct 60% of the time. Further information is in the following table.

1. What is the annualized active risk of Soi's sector rotation strategy?
2. What is the expected annualized active return of Soi's sector rotation strategy?
3. What will be the allocation to the consumer discretionary sector if Soi feels that consumer staples will outperform the consumer discretionary sector over the next month and if the active risk is limited to 5.20%?

Sector	E(R)	σ	Benchmark weight
Consumer staples	10.8%	3.0%	65%
Consumer discretionary	13.2%	5.0%	35%

[LOS 6.e] Compare active management strategies, including market timing and security selection, and...

2.

Sector rotation

Example: Sector rotation

Answer:

$$1. \text{ Monthly active risk} = \sigma_C = \sqrt{\sigma_X^2 - 2\sigma_X\sigma_Yr_{XY} + \sigma_Y^2}$$

$$= \sqrt{0.03^2 - 2(0.03)(0.05)(0.30) + 0.05^2}$$

$$= 0.05 \text{ or } 5\%$$

$$\text{Annualized active risk} = \sigma_A = \sigma_C \cdot \sqrt{BR}$$

$$= 0.05 \times \sqrt{12} = 0.1732 \text{ or } 17.32\%$$

$$2. \text{ We have } IC = 2(\% \text{ correct}) - 1 = 2 \times 0.60 - 1 = 0.20$$

$$\text{Annualized active return} = IC \cdot \sqrt{BR} \cdot \sigma_A$$

$$= 0.20 \times \sqrt{12} \times 0.1732$$

$$= 0.12 \text{ or } 12\%$$

3. If active risk is limited to 5.20%, the deviation from the benchmark weights of 65% and 35% is limited to $(5.20\% / 17.32\%) = 30\%$.

When Soi feels that consumer staples will outperform, the allocation to that sector will be 65% + 30%, or 95%, and the allocation to consumer discretionary will be 5%.

[LOS 6.f] Describe the practical strengths and limitations of the fundamental law of active management

1.

Strengths and limitations of the fundamental law

Strengths

The fundamental law can be used to evaluate a range of active strategies, including security selection, market timing, and sector rotation.

Limitations

Garbage in, garbage out problem: Poor input estimates lead to incorrect evaluations.

In the case of unconstrained optimization, the *two* components (inputs) that determine the information ratio are:

- (1)** The information coefficient (IC), and
- (2)** The breadth (BR) of the manager's strategy

The limitations are generally derived from inaccurate estimates of the *two* inputs:

- (1)** Ex-ante measurement of skill, and
- (2)** Independence of investment decisions

[LOS 6.f] Describe the practical strengths and limitations of the fundamental law of active management

1.

Strengths and limitations of the fundamental law

01

Ex-ante measurement of skill

The IC is an estimate of the accuracy of an active manager's forecasts on an *ex-ante* basis.

One problem: that managers tend to *overestimate* their ability to outperform the market

→ *Overestimate* their IC.

02

Independence of investment decisions

The BR is the number of *truly independent* decisions that an active manager makes.

One problem: two or more decisions rely on same (or similar) information → They are *not* independent.

If individual decisions are correlated, the breadth can be estimated as:

$$BR = \frac{N}{1 + (N - 1)r}$$

where N = number of decisions

r = correlation between the decisions

Limitations

[LOS 6.f] Describe the practical strengths and limitations of the fundamental law of active management

1.

Strengths and limitations of the fundamental law

Limitations

02

Independence of investment decisions

Practical problem: Decision independence may be compromised by systemic influences within a strategy, the cross-sectional dependency.

Example: A value strategy applied to different stocks within an industry may not be truly independent (most stocks will have similar fundamentals, such as P/E ratio)

Practical problem: Decision independence can be compromised by time-series dependency.

Example: Monthly rebalancing decisions may not be truly independent from period to period.